

Beggar-Thy-Self: National Standing and the Politics of Environmental Protectionism in Differentiated Markets

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Abstract

This paper presents a model on the political economy of environmental policies in the context of two trading countries, each with one firm which acts as Cournot duopolists producing differentiated products. Both the domestic and the foreign firm producing polluting goods lobby for lax environmental regulations to achieve a competitive edge in the international market. The incumbent's goal in each country is to be re-elected to power and he/she balances both social welfare and campaign contributions from the lobby while setting the optimal environmental policy. The results show that higher product differentiation leads to more stringent regulations, even under lobby pressure, contrary to the standard deregulatory effect of firm lobbying. High product differentiation tempers the domestic firm's burden of stricter regulations by shielding the market share from its rival. However, this standard relationship between product differentiation and stringency in environmental regulations reverses when the incumbent's payoff embeds a psychological cost for a benchmark-exceeding import level, linked to sovereignty and domestic market dominance. Given the rival's policy, the duopoly setup *à la Cournot* incentivises the incumbent to lower taxes to restrict imports. Since every incumbent across countries is individually rational, operating under identical incentives in a symmetric equilibrium, any unilateral tax cut by the incumbent becomes an inefficient instrument to stifle import penetration as every incumbent cuts taxes competitively. The effect of the tax cut abroad on home imports dominates the effect of the domestic tax cut, which causes a protectionist paradox that expands, rather than suppresses, foreign market penetration. This self-defeating effect amplifies as products become highly differentiated, resulting in lower cross-price elasticities between domestic and foreign goods. In order to achieve even a moderate reduction in imports, the incumbent is forced to pursue increasingly aggressive deregulation.

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1 Introduction

Environmental externalities associated with industries producing differentiated products have been a focal point of discussion with the surge in trade in imperfect markets. Sectors such as automotive, electronics, textiles and chemicals, which produce differentiated products, contribute significantly to global pollution. For instance, the report by the Corporate Climate Responsibility Monitor 2025 posits that the automotive sector contributes to around 20 per cent of global green house gas emissions. From an environmental policy standpoint, it becomes crucial to study such sectors as markets get more interconnected and integrated owing to globalisation and trade. With a gradual transition in the landscape of international trade towards trade within industries producing differentiated products, the scope of strategic interaction between governments alter the dynamics of how (sub)optimal environmental policies are chosen.

The trade-environment literature is replete with studies on how trade liberalisation deteriorates environmental quality as countries compete to achieve a competitive edge over their trading partners (Barrett, 2001; Kennedy, 1994). In contrast, it has been well established that nations can use stricter environmental standards to restrict trade when they are constrained by international treaties and trade agreements (Pfluger, 2001). Studies that juxtapose the implications of environmental regulations and their linkages with trade policies in an oligopolistic framework include Conrad (1993), Barrett (2001), Ulph and Ulph (1996), Walz and Wellisch (1997), Neary and Leahy (2000), Conrad (2001), Tanguay (2001), Burguet and Sempere (2003), Jinji (2005). A broader conclusion of these studies is that unlike weaker environmental regulations, export subsidies are welfare improving. Weaker environmental regulations are typically costly to the society as they worsen the environmental quality. Export/production subsidies, on the other hand, are not costly to the society since they redistribute resources from the government to industries/firms. Apart from emission taxes as potential policy instruments to control for externalities from pollution, subsidies on pollution abatement and investment credits on cleaner technologies may improve international competitiveness of domestic firms (Conrad, 1993, 2001; Neary and Leahy, 2000). On the contrary, a recent study by Buccella et al. (2025) show that emission taxes are welfare-improving as opposed to abatement subsidies as the gains from the decrease in environmental damage and higher government tax revenue outweigh the negative effect on firm profits and consumer surplus when there exists inefficient abatement technology. These studies broadly consider differences in abatement efforts, asymmetric costs, and differences in pollution intensity of firms as factors determining optimal environmental tax which may be set above or below the social marginal damage.

Although most of these studies focus mainly on homogeneous products in a Cournot duopoly setting, product differentiation is a crucial aspect in setting optimal environmental policies (Espinola-Arredondo and Zhao, 2012; Fujiwara, 2009; Gautier, 2013; Katsoulacos and Xepapadeas, 1996; Lahiri and Symeonidis, 2007; Lambertini, 2013). Fujiwara (2009), Espinola-Arredondo and Zhao (2012), and Lambertini (2013) assumes horizontal product differentiation and shows that for an exogenous num-

ber of oligopoly firms optimal pollution tax is positive when the damage from pollution is large enough. This is in sharp contrast to the result in [Katsoulacos and Xepapadeas \(1996\)](#), where the short run emission tax is less than the marginal social damage. [Fujiwara \(2009\)](#) also extends the short run analysis to the long run where tax rates are set high to restrict entry of firms when products are homogeneous and there is no gain from variety. The tax instrument addresses two distortions: pollution damage and excess entry. [Gautier \(2013\)](#) studies how governments may strategically choose environmental tax and abatement subsidy with restricted market entry, which is an extension of the study by [Lahiri and Symeonidis \(2007\)](#) where firms face cost asymmetries. Cost asymmetries play a vital role in increasing differentiated markets where the firm's objective also lies differentiating itself in terms of environmental consciousness [Gautier \(2017\)](#).

Another strand of literature on polluting oligopoly firms incorporate the role of green lobbies in addressing inefficiencies in environmental policy choices ([Nakada, 2020](#)). Contrary to common belief, environmentalists may alter the political economy of environmental taxes by favouring deregulation to reduce foreign pollution ([Aidt, 2005](#); [Canton, 2008](#)). This happens when the actual marginal environmental damage at home is less than the perceived marginal environmental damage abroad. In addition to green lobbies, studies also highlight the role of producer lobbies in determining environmental regulations. [Bontems et al. \(2025\)](#) highlight the relative strength of green lobbies in comparison to producer lobbies as a key determinant of the relative stringency of optimal environmental tax. While the commonly held belief is that the interest of producer lobbies is to exert political pressure to reduce environmental regulations ([Anger et al., 2015](#); [Pacca et al., 2021](#)), heterogeneous abatement technologies and production processes and asymmetric costs may alter the outcome ([Cai and Li, 2020](#); [Kalk and Sorger, 2023](#)). Typically, the political pressure from the cleaner firms favour stringent regulations as opposed to the dirty sectors owing to the comparative advantage embedded in lower abatement costs. The strategic interaction between the cleaner and the dirty firm results in lobbying competition where the clean firm lobbies for stringent regulation and the dirty firm opposes the move.

The contribution of the present study lies at the intersection of the studies by [Fujiwara \(2009\)](#) and [Bontems et al. \(2025\)](#). Unlike [Fujiwara \(2009\)](#), the present model considers a two-country open economy trading in differentiated products produced by the Cournot duopoly firms, one in each country. Production of each variety of the differentiated good leads to transboundary externalities. The model also introduces firm lobbies that have stakes in the market share and create political pressure on the incumbent government to lower environmental regulations to leverage the cost advantage. The study can be positioned closely with [Bontems et al. \(2025\)](#) with homogeneous goods as it contributes to this literature by incorporating a differentiated oligopoly market linked with trade and environmental externalities. This model, however, assumes away the presence of green lobbies since the mechanism of interest here is the producer's political pressure, and not the environmentalist's.

Another novel contribution of this study lies in modelling import aversion of the incumbent government as one of the determinants of the overall payoff. Typically, protectionist governments place

import aversion to relative differences in domestic economic outcomes over some salient foreign benchmark. This draws from the literature on envy and malice where economic agent's subjective well-being is influenced not only by own income or consumption but also on income and consumption of others around them. Utility and payoff comparisons across individuals affect the equilibrium policy choices. Such policies would typically be shaped by the relative differences in national incomes across countries, especially when global markets are open to trade (Mayda and Rodrik, 2005). Skaperdas and Testa (2025) go a step further by introducing psychological payoff associated with group status, which is tied to the relative success of the nation, measured by the country's national income. This mirrors the inherently comparative aspect of group status as discussed by Sambanis and Shayo (2013). In other words, protectionist sentiments stem from the status payoff, which is driven by the relative income/output of the nation in terms of achievements and prestige ascribed to provisioning of public goods such as high-speed railways, path-breaking discoveries, space missions, and so on. Moreover, the intensity of public investments may not necessarily reflect economic efficiency when governments systematically lean towards cultivating national salient status-quo. For example, Ng (1987) finds that the optimal level of expenditure on public goods is higher when relative income comparisons are incorporated into individual welfare. Such income comparisons play a crucial role in policy choices by governments (Chaudhry and Garner, 2013). Mui (1995), Carroll et al. (1997), and Alvarez-Cuadrado et al. (2004) incorporate income comparisons and envy in models of economic growth and highlight their role in providing incentives to the government and incumbents in a optimal policy choices such as the incidence of innovation, and capital accumulation. These predictions merit empirical credence in the studies by Easterlin (1995) and Luttmer (2005) who find that individuals of households with higher income/utility are happier but there exists a negative effect of increases of neighbour's income/utility on self-reported happiness. Among historical examples, Lie (1998) points out that South Korea recognised Japan was, by decades, ahead of South Korea after the Korean War. This was manifested in protectionist sentiments and a sense of inferiority to 'catch-up'.

Drawing from the literature on salient status-quo and psychological payoff including Mayda and Rodrik (2005) and Akerlof and Kranton (2000), I incorporate a psychological cost associated with any deviation from an salient status-quo-consistent outcome for the incumbent. The context is anchored in the degree of protectionism and the pride placed on domestic market dominance (or aversion to foreign market penetration), which is politically salient. The incumbent government attaches strong sentiments to domestic strength over foreign competition. This provides an additional layer of policy distortion beyond the standard canonical framework on protectionist policies in the presence of special interest groups (Bombardini, 2008; Conconi, 2003; Fredriksson, 1999; Gawande and Bandyopadhyay, 2000; Grossman and Helpman, 1994,9; Maggi and Rodriguez-Clare, 1998). The aspect of import aversion adds a unique dimension to protectionism that traverses the economic factors such as the use of 'ostensible' measures imposed on trade flows citing human safety and environmental concerns. However, import aversion-driven environmental measures may not achieve the intended outcome, as all

trading partners operate under identical motives and incentives in a symmetric equilibrium. Moreover, this effect is amplified as product differentiation rises, which depresses the cross-price elasticity between domestic and foreign goods. Therefore, leveraging regulatory measures like tax cut to restrict foreign penetration become inefficient as even a moderate level of import reduction demands more aggressive deregulation among trading partners. Product differentiation, which acts as a facilitator of environmental stringency in the baseline scenario, absent the incumbent's psychological cost, undergoes a complete reversal under import-aversion where it acts as a catalyst for deeper deregulation.

The present study develops two benchmark scenarios, namely, the regime of the cooperative social planner and the non-cooperative strategic trade regime, before introducing political lobbying and import aversion. The global social planner always sets a strictly positive environmental tax independent of the share of environmental damage as the reallocation changes and not the total damage. The non-cooperative equilibrium accounts for both environmental damage and terms-of-trade rents from foreign consumers. With low environmental spillovers, high product differentiation shrinks the elasticity of demand for foreign consumers, inducing the incumbent to set a higher environmental tax relative to the cooperative benchmark policy. High pollution spillovers are met with weaker regulations relative to the cooperative benchmark as stricter regulations lead to emission leakages

The present work, therefore, contributes to the literature on the political economy of environmental policy by incorporating the link with trade in differentiated products and transboundary pollution in a Cournot duopoly framework where the incumbent government faces aversion to foreign penetration in a unified framework. The results show that the incumbent government may choose a more stringent environmental tax even in the presence of producer lobbies who demand less regulation to reduce costs of production. This is in contrast to the standard result where incumbents set laxer environmental regulations to meet the demands of producer lobbies ([Aidt, 1998](#); [Barbier et al., 2005](#); [Damania et al., 2003](#); [Grey, 2018](#); [Welsch, 2004](#)). In the present study, producer lobbies do not oppose stringent environmental regulations when their products are highly differentiated, and this result holds with identical cost structures across firms without any abatement technology unlike [Kalk and Sorger \(2023\)](#), [Cai and Li \(2020\)](#) and [Grey \(2018\)](#). In addition to the interaction with the producer lobby, the incumbent's environmental policy may also be driven by non-economic factors such as national salient status-quo import levels may lead to further reduction in equilibrium environmental tax, aggravating the problem of environmental damage.

This explanation, however, is contingent upon a fixed environmental policy of the rival. Since the incentive to cut taxes to restrict imports and to shield the domestic firm from foreign competition is identical across trading countries in a symmetric equilibrium, the policy backfires. As both countries cut taxes competitively, import penetration overshoots the level that would prevail in the absence of the psychological cost. The effect of a tax cut in the foreign country on home imports dominates the effect of the domestic tax cut, resulting in a self-defeating impact, which boosts import penetration rather than restricting it. Higher product differentiation exacerbates this paradoxical effect owing to low cross-

price elasticity between domestic and foreign goods. Even a moderate reduction in imports requires incumbents in both countries to pursue aggressive deregulation.

The paper is organised as follows. [Section 2](#) discusses model setup followed by [Section 3](#), which covers the baseline scenario of cooperative equilibrium. [Section 4](#) discusses the non-cooperative strategic trade equilibrium. [Section 5](#) covers the political equilibrium. [Section 6](#) is an extension of the political lobbying model that incorporates the psychological cost of the incumbent government ascribed to foreign penetration. [Section 7](#) concludes.

2 The Model

2.1 Setup

I consider a model of reciprocal market comprising two countries, namely Home and Foreign, and two firms H and F, respectively that act as duopolists in the international market and compete in output levels. Each firm produces a horizontally differentiated variety of a traded good. Here, firm H produces x for the home market and exports x^* to the foreign market. Total production by firm H is $X = x + x^*$. Similarly, the home country imports y from Foreign, which is the surplus from total production of Y after meeting the domestic market demand y^* in Foreign. I abstract from transportation costs associated with trade and normalise them to zero.

I assume that the two firms share the same technology and do not incur any fixed production costs. The marginal cost of production, $c \in (0, 1)$ is identical across firms. Production of the differentiated good is polluting such that one unit of production emits one unit of emission that is harmful to the environment. Each firm bears the tax burden on emissions, and abatement activities by firms are assumed away.

Apart from the duopoly firms, both Home and Foreign economies produce a homogeneous good (numeraire) z and z^* , respectively, using labour alone in a perfectly competitive market. The production exhibits constant returns to scale, and by appropriate choice of units, the input coefficient is assumed to be unity. Under the assumption of perfect competition, wage rate in the homogeneous goods sector is internationally fixed to unity. The aggregate supply of labour is sufficiently large to maintain a positive level of output in equilibrium.

Both Home and Foreign are identical in all respects, with foreign country variables being denoted by an asterisk (*). For brevity, the analysis will primarily focus on the home country.

2.2 Consumers

A representative consumer at Home is characterised by a quadratic and quasi-linear utility function

$$U = c_z + u(c_x, c_y) - \xi, \quad (2.1)$$

207 where

$$u(c_x, c_y) = (c_x + c_y) - \frac{1}{2}(c_x^2 + c_y^2 + 2\sigma c_x c_y)$$

208 . c_z, c_x, c_y denote consumption of goods z, x , and y respectively. $\sigma \in [0, 1]$ ¹ measures product differentiation such that in the limiting case as $\sigma \rightarrow 0$, x and y become highly differentiated and independent.
 209
 210 On the other hand, as $\sigma \rightarrow 1$, x and y become homogeneous (perfect substitutes). The utility function is
 211 differentiable, increasing and concave in its arguments. The term ξ captures the environmental damage
 212 from production of the polluting goods, and enters as disutility to the representative consumer. The
 213 damage function is a weighted average of emissions in the domestic and the foreign country from the
 214 production of the differentiated goods.

215 The emissions from production of home variety ($e(x, x^*)$) and foreign variety ($e^*(y, y^*)$) are
 216 quadratic in nature, such that,

$$e(x, x^*) = \frac{1}{2}(x + x^*)^2, \quad (2.2)$$

217 and

$$e^*(y, y^*) = \frac{1}{2}(y + y^*)^2. \quad (2.3)$$

218 Here, $e_i > 0, e_{ii} \geq 0, e(0) = 0$ for $i = x, x^*$. The same holds true for $e^*(y, y^*)$.

219 Therefore the environmental damage function at Home is given by

$$\xi(x, x^*, y, y^*) = (1 - \theta)e(x, x^*) + \theta e^*(y, y^*) = \frac{1}{2}(1 - \theta)(x + x^*)^2 + \frac{1}{2}\theta(y + y^*)^2. \quad (2.4)$$

220 The parameter $\theta \in [0, 1]$ is the relative weight placed on foreign emissions, which is carried across the
 221 border by wind or water. When pollution is global in nature, $\theta = \frac{1}{2}$. For local pollution, $\theta = 0$. For
 222 $\theta \in (0, \frac{1}{2})$, there exists regional environmental problems. The damage function $\xi(\cdot)$ obeys the standard
 223 properties: it is increasing and convex in emissions.

224 Similarly, for the Foreign country, environmental damage is given by

$$\xi^*(x, x^*, y, y^*) = \frac{1}{2}(1 - \theta)(y + y^*)^2 + \frac{1}{2}\theta(x + x^*)^2. \quad (2.5)$$

225 The budget constraint faced by a representative consumer at Home is given as

$$pc_x + qc_y + c_z = m, \quad (2.6)$$

226 where p, q denote domestic prices of x and y , respectively and m denotes income.

227 Utility maximisation yields inverse demand functions as

$$p = 1 - c_x - \sigma c_y, \quad (2.7)$$

¹The parametric restriction for σ may be assumed as $\sigma \in (0, 1)$ to explore the interesting scenarios unlike cases of perfect substitutes and independent/unrelated demand for domestic and foreign variety of the good.

and

$$q = 1 - c_y - \sigma c_x. \quad (2.8)$$

Consumer surplus is given by

$$S = u(c_x, c_y) - p(c_x, c_y)c_x - q(c_x, c_y)c_y - \xi. \quad (2.9)$$

Substituting the inverse demand functions from (2.7) and (2.8) and incorporating the assumption that the market clearing conditions are met (production matches demand) in (2.9) yields

$$S(x, y) = \frac{1}{2}(x^2 + y^2) + \sigma xy - \left[(1 - \theta) \frac{1}{2}(x + x^*)^2 + \theta \frac{1}{2}(y + y^*)^2 \right]. \quad (2.10)$$

Similarly, consumer surplus in Foreign is given by

$$S^*(x^*, y^*) = \frac{1}{2}(x^{*2} + y^{*2}) + \sigma x^* y^* - \left[\theta \frac{1}{2}(y + y^*)^2 + (1 - \theta) \frac{1}{2}(x + x^*)^2 \right], \quad (2.11)$$

where $\theta \in [0, 1]$ is the weight placed on emissions at Home.

This completes the consumption side.

2.3 Producers

The numeraire good Z is produced using homogeneous labour input L with constant returns to scale in a perfectly competitive market setting. By appropriate choice of units, the labour-output coefficient is assumed to be unity, that is, one unit of labour produces one unit of output Z , with wage rate equal to unity. The aggregate labour supply is sufficiently large to ensure a positive level of output of Z . There is no pollution associated with production or consumption of Z . The differentiated good, $\tilde{X}$, is produced with constant marginal cost $c \in (0, 1)$. Pollution damages which arise from the production of X and X^* are subject to a charge imposed by the policy maker, say a pollution tax.

Profit of the home firm is given as

$$\pi = p(x, y)x + p^*(x^*, y^*)x^* - c(x + x^*) - t(x + x^*). \quad (2.12)$$

The home firm sells x in the home market at price p and exports x^* to Foreign at price p^* . It incurs marginal cost c and pays an environmental tax t per unit of production to the Home government.

For the foreign firm which exports y at price q and sells y^* in Foreign at price q^* , the profit function is given as

$$\pi^* = q^*(x^*, y^*)y^* + q(x, y)y - c(y + y^*) - t^*(y + y^*), \quad (2.13)$$

where c is the marginal cost and t^* is the per unit environmental tax paid to the Foreign government from production of variety Y .

Both Home and Foreign firms compete *à la* Cournot in the two countries and choose equilibrium outputs by maximising profits, given the tax rates t and t^* . The equilibrium outputs of each firm are given as

$$x = x^* = \frac{(1 - c)(2 - \sigma) - 2t + \sigma t^*}{4 - \sigma^2}, \quad (2.14)$$

and

$$y = y^* = \frac{(1 - c)(2 - \sigma) - 2t^* + \sigma t}{4 - \sigma^2}. \quad (2.15)$$

Using the first-order conditions for profit maximisation, equilibrium profit of the Home and the Foreign firm can be expressed as

$$\pi = x^2 + x^{*2}, \quad (2.16)$$

and

$$\pi^* = y^2 + y^{*2}, \quad (2.17)$$

respectively.

2.4 Environmental Tax Revenue

The government in each country collects revenue from the environmental tax on domestic production and distributes the revenue to the consumer in a lump sum fashion. I define revenue as

$$R(t; t^*) = t(x + x^*), \quad (2.18)$$

in the home country, and

$$R^*(t^*; t) = t^*(y + y^*), \quad (2.19)$$

in the foreign country.

It is important to note here that the markets are segmented, that is, each firm considers each country as a separate market and makes distinct output decisions for each. The duopoly firms act as rivals in the world market and try to extract profits by capturing a greater market share. Policies such as environmental tax are meant for regulating domestic output by internalising the negative externality. Therefore, home pollution tax affects domestic production meant for home consumption as well as exports to the foreign country. Similarly, foreign pollution tax affects the output of the foreign firm which is consumed within its territories as well as exports to the home country.

[Appendix A](#) provides the marginal effects of policy change.

3 Global Social Planner: Cooperative Equilibrium

The normative benchmark in this model is a scenario where the benevolent social planner maximises the joint social welfare function, which is a sum of aggregate welfare functions of Home and Foreign. There is no presence of lobby in the determination of environmental tax in both the countries.

Each country's welfare is the sum of labour income, consumer surplus, domestic firm profit, and environmental tax revenue net of environmental damage. The home country's welfare is given by

$$G(t; t^*) = l + S(t; t^*) + \pi(t; t^*) + R(t; t^*). \quad (3.1)$$

The foreign analogue is given by

$$G^*(t^*; t) = l^* + S^*(t^*; t) + \pi^*(t^*; t) + R^*(t^*; t). \quad (3.2)$$

The welfare functions $G(t; t^*)$ and $G^*(t^*; t)$ are well-behaved. They are continuous over a non-empty, closed, bounded and convex set of the policies t and t^* , respectively. They are twice differentiable and concave in the policies t and t^* , respectively. This ensures a unique interior solution, along with the concavity condition which yield a maximum.

The joint social welfare function, $\Omega(t, t^*)$ is defined as

$$\Omega(t, t^*) = G(t, t^*) + G^*(t^*, t). \quad (3.3)$$

The global social planner chooses a common tax rate such that $t = t^*$ to maximise $\Omega(t, t^*)$, internalising the transboundary externality across countries.

Using (2.10), (2.11), and (2.14)-(2.19), the equilibrium environmental tax under cooperation is given by

$$t^{SW} = t^{*SW} = \frac{1-c}{3+\sigma} > 0. \quad (3.4)$$

This result can be stated as

Proposition 1. *The global social planner chooses a strictly positive environmental tax under cooperation.*

Proof. Differentiating $\Omega(t, t^*)$ with respect to a common tax $t (= t^*)$ gives the equilibrium tax for the global social planner. t^{SW} is always positive for every $\sigma \in [0, 1]$ and $c \in (0, 1)$. ■

This implies that the social planner does not subsidise emissions. The equilibrium tax rate is independent of the environmental damage parameter θ . The exogenous parameter θ is the spillover share of environmental damage attributed across two countries. As a result, it does not change the total global environmental damage. The social planner's joint welfare function does not include θ as it cancels out

while summing up each country's share of environmental damage. This leads to an environmental tax that depends only on c and σ .

4 Non-cooperative Equilibrium with Strategic Trade

In this section, I derive the non-cooperative equilibrium under strategic trade where lobbies are absent. This can also be interpreted as a scenario where the incumbent government values only the welfare of the general voters and does not give in to the producer lobby's demand for deregulation.

The simultaneous-form game is solved through backward induction. In stage 1, the Home and the Foreign governments choose the environmental policy non-cooperatively and simultaneously, assuming their rival's policy as given. Using the tax rates set by the respective government, each firm simultaneously chooses the Cournot quantity, holding the rival's output as given.

The incumbent government at Home chooses an environmental tax by maximising the welfare function

$$G(t; t^*) = l + S(t; t^*) + \pi(t; t^*) + R(t; t^*). \quad (4.1)$$

Using (2.10)-(2.11), (2.14)-(2.19), the equilibrium environmental tax under non-cooperation is given by

$$t^{ST} = t^{*ST} = \frac{(1-c)N(\sigma, \theta)}{D(\sigma, \theta)}, \quad (4.2)$$

where $N(\sigma, \theta) = \sigma^2 + (4\theta + 1)\sigma + (8\theta - 6)$ and $D(\sigma, \theta) = \sigma^2 + (4\theta - 3)\sigma + (8\theta - 14)$.

Proposition 2. Under the second-order condition $D(\sigma, \theta) < 0$ and $c \in (0, 1)$, the national welfare maximising incumbent government chooses an environmental tax, ($t^{ST} > 0$) in the equilibrium if and only if

$$\theta < \theta_{ST}(\sigma) \equiv \frac{(3+\sigma)(2-\sigma)}{4(2+\sigma)}. \quad (4.3)$$

$\forall \theta > \theta_{ST}$, the equilibrium policy is an environmental subsidy ($t_{ST} < 0$).

Proof. Given $c \in (0, 1)$ and the second-order condition $D(\sigma, \theta) < 0$, the sign of the equilibrium policy depends on the sign of $N(\sigma, \theta) = \sigma^2 + (4\theta + 1)\sigma + (8\theta - 6)$, which is monotonically increasing in θ given the value of $\sigma \in [0, 1]$.

Setting $N(\sigma, \theta) = 0$, yields the threshold value $\theta_{ST}(\sigma) = \frac{(3+\sigma)(2-\sigma)}{4(2+\sigma)}$.

Sufficiency condition. If $\theta < \theta_{ST}(\sigma)$ and with $D(\sigma, \theta) < 0$ and $c \in (0, 1)$, the equilibrium policy is an environmental tax ($t^{ST} > 0$).

Necessary condition. If $t^{ST} > 0$, and with $D(\sigma, \theta) < 0$ and $c \in (0, 1)$, requires $N(\sigma, \theta) < 0$. Since the function $N(\sigma, \theta)$ is monotonically increasing in θ , it implies $\theta < \theta_{ST}(\sigma)$. ■

The non-cooperative strategic trade outcomes are driven by the trade-off faced by the national government: environmental quality and market power. When θ is very small ($\theta < \theta_{ST}(\sigma)$), pollution is more local as $1 - \theta$ is very high and less transboundary pollution enters the home country. As a result, the national government sets an environmental tax to clean up the local emissions. This adversely affects the domestic firm's profit. However, the tax restricts domestic output that leads to a rise in the world price. Foreign consumers pay higher prices for the exports from the home country. The gain from terms-of-trade and higher tax revenue outweighs the loss of market power of the domestic firm. When σ is very low, the home producer does not necessarily lose market share as the products are highly differentiated with a low elasticity of demand for foreign consumers. So the terms-of-trade gain and improvement in environmental quality from a tax exceeds the loss in the firm's profit.

As the share of transboundary pollution rises above the threshold ($\theta > \theta_{ST}(\sigma)$), the domestic government's incentive to tax pollution falls. A lot of pollution is drifted in from the foreign country. In this scenario, cutting down domestic emissions hardly affects the total emissions. Taxing emissions at Home reduces home output and home firm's profit, which simultaneously expands foreign output and foreign firm's profit under the Cournot set-up. This invariably leads to emission leakage into the home country. Consequently, the environmental considerations compel the home government to subsidise emissions. Here, the improvement in environmental quality from a tax is outweighed by the cost of the home firm's losing its market power and profit [see [Appendix B](#) for detailed proof].

Comparing the equilibrium tax rates under cooperation and non-cooperation in the absence of producer lobby yields

$$t^{SW} - t^{ST} = \frac{(1-c)(2+\sigma)^2(1-4\theta-\sigma)}{(3+\sigma)[\sigma^2 + (4\theta-3)\sigma + (8\theta-14)]} \quad (4.4)$$

Proposition 3. *The global social planner's tax exceeds the strategic trade and non-cooperative tax ($t^{SW} > t^{ST}$) tax equilibrium if and only if $\theta < \frac{1-\sigma}{4}$.*

(i) *If $\theta > \frac{1-\sigma}{4}$, then $t^{SW} > t^{ST}$,*

(ii) *If $\theta < \frac{1-\sigma}{4}$, then $t^{SW} < t^{ST}$, and*

(iii) *If $\theta = \frac{1-\sigma}{4}$, then $t^{SW} = t^{ST}$.*

Proof. Comparing the two scenarios, one with cooperation and the other with non-cooperation with strategic trade shows that the magnitude of the taxes differ by the value of θ .

The difference between the two taxes is given by

$$t^{SW} - t^{ST} = \frac{1-c}{3+\sigma} - \frac{(1-c)N(\sigma, \theta)}{D(\sigma, \theta)}. \quad (4.5)$$

Simplifying (4.5) yields

$$t^{SW} - t^{ST} = \frac{(1-c)(2+\sigma)^2(1-4\theta-\sigma)}{(3+\sigma)D(\sigma, \theta)}. \quad (4.6)$$

Under the parametric restrictions $c \in (0, 1)$, $\sigma \in [0, 1]$, and the second-order condition for maximum ($D(\sigma, \theta) < 0$), the sign of (4.6) depends only on the last term of the numerator. Considering the last term of the numerator, one can find the threshold value of the damage parameter θ .

The threshold value of θ is obtained from

$$1 - 4\theta - \sigma = 0 \implies \tilde{\theta}(\sigma) = \frac{1 - \sigma}{4}, \quad (4.7)$$

where $\tilde{\theta}(\sigma)$ is the threshold value of θ .

Therefore,

$$t^{SW} - t^{ST} = \begin{cases} > 0, & \text{if } \theta > \bar{\theta}, \\ 0, & \text{if } \theta = \bar{\theta}, \\ < 0, & \text{if } \theta < \bar{\theta}. \end{cases} \quad (4.8)$$

The sufficient condition for $t^{SW} > t^{ST}$ is given by $\theta > \bar{\theta} = \frac{1 - \sigma}{4}$, which yields $1 - 4\theta - \sigma < 0$. This implies $t^{SW} > t^{ST}$ given that $(1 - c) > 0$, $(2 + \sigma)^2 > 0$, $3 + \sigma > 0$, and $D(\sigma, \theta) < 0$. Similarly, $\theta < \bar{\theta} = \frac{1 - \sigma}{4}$ implies $t^{SW} < t^{ST}$. When $\theta = \bar{\theta} = \frac{1 - \sigma}{4}$, $t^{SW} = t^{ST}$.

The social planner cares only about the total environmental damage unlike national government who sets a tax on the environmental damage that affects its own country. ■

The above proposition states that the non-cooperation between Home and Foreign incumbents in (a Cournot set-up) leads to strategic substitution in terms of output and environmental damage. For $\theta > \bar{\theta}$, the share of Home's environmental damage is lower. Any unilateral rise in t^{ST} , forces domestic production and profits to fall leading to an expansion of the foreign firm's output and profit. This results in a leakage of foreign pollution to the home country. As a result, it dampens the incentive of the Home government to regulate emissions at Home. The global social planner, on the other hand, cares about the total global environmental damage, and chooses a strictly higher tax ($t^{SW} > t^{ST}$). This in turn drives the emission leakage to zero.

On the contrary, the optimal tax under strategic trade exceeds the optimal tax set by the global social planner when environmental damage is low, $\theta < \bar{\theta}$. Lower θ invokes the national government to prioritise terms-of-trade motive over environmental quality. It raises the environmental tax to restrict domestic output that leads to a rise in the world price. This draws monopoly rents from foreign consumers. The gain from terms-of-trade and higher tax revenue outweighs the loss of market power of the domestic firm. The market power of the home firm stems from σ . Lower is the value of σ , the incentive to extract foreign rents rises as products are highly differentiated with low demand elasticity of foreign consumers. The global social planner maximises global welfare and considers the loss in consumer surplus of the foreign consumers while setting the global environmental tax. Since $\theta < \frac{1 - \sigma}{4}$, the global social planner weighs consumer surplus more than the low environmental damage. As higher taxes lead to a rise in prices, the social planner chooses a lower tax to make goods affordable to consumers.

When $\theta = \bar{\theta}$, the rent extraction and terms-of-trade motive of the national government offsets the motive of internalising the environmental damage. The optimal environmental tax under non-cooperation coincides with the globally efficient tax set by the social planner ($t^{SW} = t^{ST}$).

This completes the analysis of the benchmark policies with no producer lobbies. The next section discusses the equilibrium environmental policies when producer lobbies exist and the policy maker values campaign contributions by the lobby.

5 The Political Model

It is widely established in the political economy literature that governments/incumbents care about office and the prospects of being re-elected to power, which is determined by (Arulampalam et al., 2009) the average welfare of the citizenry. In democracies, the overall popularity of the incumbent government and its policy stance play a key role in electoral victory. As a result, the goal of the incumbent government is to choose policies that maximise the total welfare comprising producer surplus, consumer surplus net of environmental damage, and government revenue from environmental policies. While this aspect assumes special significance for the incumbent's likelihood of being re-elected to power, he/she may also be influenced of certain special interest groups through *quid pro quo* arrangements such as sale of (de)regulation to serve the groups whose payoffs are contingent on the policies chosen. Consequently, the incumbent government weighs both contributions (*bribe*) from the interest groups and welfare of the general voter while setting policies. It is assumed that the incumbent/s preference is exogenous, which is reflected in the relative weight attached to the welfare of the average voter relative to the contributions from interest groups. Furthermore, the incumbent's preference also aligns with that of the general voters, which rules out any divergence in preferences between voters and the incumbent.

5.1 The Lobby

In the present context, the government in each country sets the environmental tax subject to the influence of the producer lobbies. Since taxes raise costs of production, some owners of each firm are assumed to organise into a lobby and make campaign contributions $C(t; t^*)$ in the home country and $C^*(t^*; t)$ in the foreign country to their respective governments. Notice that campaign contributions by the lobby in each country are contingent on the environmental tax set by the respective governments, holding the policy chosen by the other government as given.

Consumers in each country have limited stakes in the political process and fail to overcome the free-rider problem to form a lobby for favourable policies.

The payoff of the producer lobby (gross of contributions) at Home and at Foreign are given by

$$W(t; t^*) = \pi(t; t^*), \quad (5.1)$$

411 and

$$W^*(t^*;t) = \pi^*(t^*;t), \quad (5.2)$$

412 respectively.

413 5.2 The Incumbent's Problem

414 As discussed earlier, the incumbent's motive is to retain political office and therefore, he/she attempts
 415 to maximise the likelihood of getting re-elected to power. The political incentives could be met through
 416 two distinct channels: *one*, campaign contributions to finance election spending; *two*, wooing the elec-
 417 torate through welfare/public good-oriented policies that strengthen political support. The latter serves
 418 as a litmus test for the incumbent government to be re-elected in the subsequent elections. A voter's
 419 well-being associated with the chosen policy would directly shape his/her voting behaviour and trans-
 420 late into the electoral outcome Villamil et al. (2021). Resultantly, the incumbent government maximises
 421 a political support function given as a weighted sum of the contributions received and the welfare of a
 422 general voter. The interaction between the incumbent government and the firm lobby can be defined
 423 as a menu auction as in Bernheim and Whinston (1986), one of which operates within each country.
 424 Therefore, each country has its own single-principal menu auction comprising one lobby and one in-
 425 cumbent government with two sub-games (one in each country) linked to the assumption of treating
 426 the rival's policy as given as in Grossman and Helpman (1995). In this context, the incumbent govern-
 427 ment can choose policies on a continuum of possible actions, which implies that the choice set of policies
 428 is continuous.

429 Following Grossman and Helpman (1995), the incumbent government at Home maximises

$$G^P = C(t;t^*) + \lambda G(t;t^*); \lambda \in (0, \infty), \quad (5.3)$$

430 where $C(t;t^*)$ denotes the contribution and λ is the weight attached to the welfare of the general voter
 431 $G(t;t^*)$ relative to the contribution received from the producer lobby. The incumbent government
 432 chooses $t \in \mathcal{T}$ assuming $t^* \in \mathcal{T}^*$ as given. The choice sets $\mathcal{T}$ and $\mathcal{T}^*$ are bounded with $\underline{t}$ and $\bar{t}$ as
 433 the upper and the lower bounds of set $\mathcal{T}$ in the home country, respectively. Similarly, $\underline{t}^*$ and $\bar{t}^*$ are
 434 the upper and the lower bounds of set $\mathcal{T}^*$ in the foreign country, respectively. The entire analysis is
 435 restricted to the interior of the policy choice sets.

436 Here, the aggregate welfare is defined as the sum of labour income, consumer surplus net of the
 437 disutility from environmental damage, profit of the Home firm, and the environmental tax revenue, that
 438 is,

$$G(t;t^*) = l + S(t;t^*) + \pi(t;t^*) + R(t;t^*). \quad (5.4)$$

Analogously, in the foreign country, the incumbent government maximises

$$G^{*P} = C^*(t^*; t) + \lambda^* G^*(t^*; t); \lambda^* \in (0, \infty), \quad (5.5)$$

where $C^*(t^*; t)$ denotes the contribution and λ^* is the weight attached to the welfare of the general voter $G^*(t^*; t)$ relative to the contribution received from the producer lobby. Here, the incumbent government chooses t^* assuming t as given.

The foreign country's aggregate welfare is similarly defined.

This setup is defined as a lobbying game with one principal (lobby) and one agent (the incumbent) in each country. The Home government faces the Home lobby and the Foreign government faces the Foreign lobby, which mirror a 'two separate single principal menu auction' (Bernheim and Whinston, 1986). Unlike multi-principal common agency problem, the present model considers only one lobby in each country, which ensures that there are no competing interests. This also rules out any cross-country coalition between Home lobby and Foreign lobby while making campaign offers/contributions to their respective agents (Home government and Foreign government, respectively). The Home sub-game and the Foreign sub-game are related through strategic interactions in policies, ruling out presence of shared principals. Incumbents in each country are Cournot players, who choose the equilibrium policy, given the policy of its rival. Each country's single principal's truthful equilibrium is a mutual best response to the other's as in Grossman and Helpman (1995) applied to trade policy. This precludes the two or multiple-principal extension of Bernheim and Whinston (1986). The coalition-proof Nash equilibrium that rests on separate truthful contribution schedules as in Bernheim and Whinston (1986), reflects a within-country auction in a one-principal-one-agent game. There is no contracting between Home lobby and Foreign lobby.

The parameter λ (and λ^*) can be interpreted as degree of honesty of the incumbent. As this value rises, the incumbent's affinity towards the aggregate welfare of the general voter rises. Social welfare takes precedence over the special interests of the lobby.

The welfare function $G(t; t^*)$ is well-behaved. It is continuous over a non-empty, closed, bounded and convex set of the policy t . It is twice differentiable and concave in the policy t . This ensures a unique interior solution, along with the concavity condition which yield a maximum. Home's (Foreign's) best-response environmental tax is a singleton for any admissible values of environmental tax in $\mathcal{T}^*$ ($\mathcal{T}$). Equilibrium environmental tax is a symmetric fixed point derived from the standard menu auction for each country separately (one-lobby-one-incumbent government for each country) as in Grossman and Helpman (1995), ruling out coalition of principals.

The political process reflects that incumbents in each country choose the policy holding the policy choice of the other as given.

5.3 Stages of the Game

The game is one-shot with simultaneous moves, which takes place in three stages and is solved through backward induction. In the first stage, the lobby chooses a contribution schedule contingent on the array of policy outcomes that maximises its payoff. In the second stage, the incumbent government sets the environmental tax conditional on the contribution schedules offered by the lobby. Neither the lobby nor the incumbent government renege on their commitments. The firm and the incumbent government in each country assume that the policy choice of their respective rival as given. Observing the set of policies announced, lobbies make contribution to the government. In the last stage, production and consumption decisions are made and all markets clear.

5.4 Non-cooperative Equilibrium

5.4.1 Determination of Campaign Contributions

Based on the menu auction framework as in [Bernheim and Whinston \(1986\)](#) and [Grossman and Helpman \(1995\)](#), (C^P, t^P) the subgame perfect Nash equilibrium satisfying following conditions:

(i) C^P is feasible;

(ii) $t^P = \arg \max_t C(t; t^*) + \lambda G(t; t^*)$;

(iii) $t^P = \arg \max_t [W(t; t^*) - C(t; t^*)] + [C(t; t^*) + \lambda G(t; t^*)]$;

(iv) $\exists a \hat{t} \in \mathcal{T}$ that maximises $C^P(t; t^*) + \lambda G(t; t^*)$ on $\mathcal{T}$ such that $C^P(\hat{t}; t^*) = 0$.

where (i) imposes a feasibility restriction on the contribution schedule of the firm lobby, which ensures that contributions are non-negative and do not exceed the aggregate payoff of the lobby; (ii) implies that the equilibrium environmental tax, t^P , maximises the incumbent's payoff function; (iii) states that the equilibrium environmental tax also maximises the joint payoff of the lobby and the incumbent; (iv) indicates that there exists a contribution of zero from the lobby to the incumbent government which the incumbent government finds equally attractive as the equilibrium policy t^P .

5.4.2 Truthful Contribution Schedules

Truthful contribution schedules reflect true preference of a lobby over the policy choice set. The net benefit of the lobby is contingent on the equilibrium contribution as a realised outcome of a set of contribution schedules. The lobby's contribution for any arbitrary policy is given by the excess of its gross welfare associated with the policy relative to some base level of welfare b , which is a scalar. Truthful contribution schedules imply that the net welfare of the lobby will be b whenever the lobby makes a positive contribution to the incumbent government in equilibrium. The lobby, therefore, attempts to make b as large as possible, which translates into paying lower $C(\cdot)$ not beyond an extent where the incumbent government deviates from the equilibrium policy t^P to some alternative policy that makes the

lobby worse-off. Suppose that the incumbent government sets a policy $\hat{t}$ when contributions are zero. The lobby can determine some b at which the incumbent government is indifferent between choosing $\hat{t}$ and t^P , that is,

$$C^O(\hat{t}, b^0; t^*) + \lambda G(\hat{t}; t^*) = C^O(t^P, b^0; t^*) + \lambda G(t^P; t^*) \quad (5.6)$$

At the baseline welfare of b^0 , the lobby makes zero contributions to the incumbent government if it sets $\hat{t}$ as the equilibrium policy. This holds when $W(t; t^*) \leq b^0$, otherwise the lobby has an incentive to raise its contribution such that the constraint of non-negative contribution binds.

More formally, a truthful contribution schedule is given by

$$C^O(t, b; t^*) = \max\{0, W(t; t^*) - b\}, \quad (5.7)$$

where b is the net welfare anchor for the lobby.

This can alternatively be expressed as the welfare of the lobby net of contributions, that is,

$$W(t; t^*) - C^O(t, b; t^*) = \min\{W(t; t^*), b\}, \quad (5.8)$$

where b is the upper bound of the net welfare.

Therefore, a truthful Nash equilibrium t^P must maximise

$$W(t; t^*) + \lambda G(t; t^*), \quad (5.9)$$

where $t \in \mathcal{T}$, and

$$C^P(t^P; t^*) + \lambda G(t^P; t^*) > C^P(t; t^*) + \lambda G(t; t^*) \quad \forall t \in \mathcal{T}. \quad (5.10)$$

If contributions are truthful, then from (5.7),

$$C^P(t^P; t^*) = W(t^P; t^*) - b^0, \quad (5.11)$$

where b^0 is the equilibrium net benefit to the lobby, and

$$C^P(t; t^*) = W(t; t^*) - b^0 \quad \forall t \in \mathcal{T}. \quad (5.12)$$

(5.10) and (5.11) together imply

$$W(t^P; t^*) + \lambda G(t^P; t^*) \geq W(t; t^*) + \lambda G(t; t^*) \quad \forall t \in \mathcal{T}. \quad (5.13)$$

The incumbent government assumes that the lobby offers a truthful contribution schedule and maximises its total payoff with a weight of $1 + \lambda$ to individuals represented by the lobby and a weight λ to the ones who do not belong to the lobby. Therefore, the welfare of the lobby enters twice in

the incumbent's payoff, once as the lobby's contribution and again through the welfare of the general voters, and hence the weight $1 + \lambda$. The general voters who do not form a lobby are considered only once in the payoff of the incumbent, and hence receive a smaller weight λ .

Assuming that the contribution schedule is differentiable around the equilibrium t^P , t^P must satisfy the following first-order conditions

$$\left[\frac{\partial W(t^P; t^*)}{\partial t} - \frac{\partial C^P(t^P; t^*)}{\partial t} \right] + \left[\frac{\partial C^P(t^P; t^*)}{\partial t} + \lambda \frac{\partial G(t^P; t^*)}{\partial t} \right] = 0, \quad (5.14)$$

and

$$\frac{\partial C^P(t^P; t^*)}{\partial t} + \lambda \frac{\partial G(t^P; t^*)}{\partial t} = 0. \quad (5.15)$$

Now (5.14) and (5.15) together imply

$$\frac{\partial W(t^P; t^*)}{\partial t} = \frac{\partial C^P(t^P; t^*)}{\partial t} \quad (5.16)$$

From (5.1), (5.16) becomes

$$\frac{\partial W(t^P; t^*)}{\partial t} = \frac{\partial \pi(t^P; t^*)}{\partial t} = \frac{\partial C^P(t^P; t^*)}{\partial t}. \quad (5.17)$$

(5.17) shows that a lobby sets its contribution such that the marginal cost of campaign funding attributed to a small change in Home's policy for any given Foreign's policy equals to the change in its gross payoff. Grossman and Helpman (1995) refers to this condition as the local truthfulness property of equilibrium contribution schedules.

Using (5.17) in (5.14) yields

$$\frac{\partial G^P(t^P; t^*)}{\partial t} = \frac{\partial \pi(t^P; t^*)}{\partial t} + \lambda \frac{\partial G(t^P; t^*)}{\partial t} = 0. \quad (5.18)$$

A similar expression is derived for the foreign country as

$$\frac{\partial G^{*P}(t^{*P}; t)}{\partial t^*} = \frac{\partial \pi(t^{*P}; t)}{\partial t^*} + \lambda^* \frac{\partial G(t^{*P}; t)}{\partial t^*} = 0. \quad (5.19)$$

The above expressions (5.18) and (5.19) are the first-order conditions for the optimisation problem faced by the incumbent government of the respective countries. Solving these conditions produces equilibrium policies t^P and t^{*P} such that $\{C^P, t^P\}$ is the best response to the policy t^{*P} and $\{C^{*P}, t^{*P}\}$ is the best response to the policy t^P . The contribution schedules must be differentiable at least around the point of equilibria.

The equilibrium t^P and t^{*P} is given by

$$t^P = t^{*P} = \frac{(1 - c)[\lambda N(\sigma, \theta) + 8]}{\lambda D(\sigma, \theta) + 8}. \quad (5.20)$$

This result can be stated as

Proposition 4. Under the parametric restrictions $c \in (0, 1)$, $\sigma \in [0, 1]$, $\lambda \in (0, \infty)$ and the second-order condition $\lambda D(\sigma, \theta) + 8 < 0$, the incumbent government chooses an environmental tax ($t^P > 0$) in the political equilibrium if and only if $\theta < \theta_P(\sigma, \lambda) \equiv \frac{(3+\sigma)(2-\sigma) - \frac{8}{\lambda}}{4(2+\sigma)}$. Equivalently, the incumbent government sets a subsidy ($t^P < 0$) whenever $\theta > \theta_P(\sigma, \lambda)$

Proof. Given $c \in (0, 1)$, $\sigma \in [0, 1]$, $\lambda \in (0, \infty)$ and the second-order condition $\lambda D(\sigma, \theta) + 8 < 0$, the sign of t^P depends on the sign of $\lambda N(\sigma, \theta) + 8$. Since $N(\sigma, \theta)$ is increasing in θ for $\lambda \in (0, \infty)$, one can find the threshold value of θ by setting $\lambda N(\sigma, \theta) + 8 = 0$, which yields

$$\theta_P(\sigma, \lambda) = \frac{(3 + \sigma)(2 - \sigma) - \frac{8}{\lambda}}{4(2 + \sigma)}. \quad (5.21)$$

The necessary and the sufficient conditions are qualitatively identical to the proof of [Proposition 2](#). ■

Corollary 1. When the incumbent government places more weight to social welfare over campaign funds ($\lambda \rightarrow \infty$), the threshold in [Proposition 4](#) converges to the no-lobby strategic trade threshold of [Proposition 3](#):

$$\lim_{\lambda \rightarrow \infty} \theta_P(\sigma, \lambda) = \theta_{ST}(\sigma). \quad (5.22)$$

Moreover, for every finite $\lambda > 0$,

$$\theta_P(\sigma, \lambda) - \theta_{ST}(\sigma) = -\frac{2}{\lambda(2 + \sigma)} < 0, \quad (5.23)$$

which implies that the political equilibrium threshold damage is strictly below the no-lobby strategic trade equilibrium. As $\lambda \rightarrow \infty$, $\theta_P(\sigma, \lambda)$ converges to $\theta_{ST}(\sigma)$.

Proof. $\lim_{\lambda \rightarrow \infty} \frac{(3+\sigma)(2-\sigma) - \frac{8}{\lambda}}{4(2+\sigma)}$, yields $\frac{(3+\sigma)(2-\sigma)}{4(2+\sigma)}$, which converges to the threshold value $\theta_{ST}(\sigma)$. The gap in $\theta_P(\sigma, \lambda)$ and $\theta_{ST}(\sigma)$ in (5.23) converges to zero as $\lambda \rightarrow \infty$. ■

(5.23) shows that the incumbent government under political influence of the producer lobby cuts down the environmental damage threshold to choose the environmental policy. A switch to environmental subsidy is governed by a strictly lower θ than the scenario when there is no lobbying. With $\lambda > 0$, the incumbent government caters to the demand from the lobby by setting a subsidy. The gap between the threshold values of θ shrinks asymptotically as the effect of political contributions on the incumbent government peters out ($\lambda \rightarrow \infty$).

Corollary 2. The equilibrium environmental tax when the incumbent government values less of political contributions ($\lambda \rightarrow \infty$) coincides with the no-lobby strategic trade outcome $t^P \rightarrow t^{ST}$ as $\lambda \rightarrow \infty$. For finite values of λ , $t^{ST} > t^P$.

567 *Proof.* Considering the political equilibrium tax when the incumbent government favours
 568 aggregate social welfare over the lobby's contribution. $\lim_{\lambda \rightarrow \infty} t^P = \lim_{\lambda \rightarrow \infty} \frac{(1-c)[\lambda N(\sigma, \theta) + 8]}{\lambda D(\sigma, \theta) + 8} =$
 569 $\frac{(1-c)N(\sigma, \theta)}{D(\sigma, \theta)} = t^{ST}$. Now, subtracting t^P from t^{ST} yields

$$t^{ST} - t^P = \frac{32(1-c)(2+\sigma)}{D(\sigma, \theta)[\lambda D(\sigma, \theta) + 8]} > 0. \quad (5.24)$$

570 since the denominator is positive and $\sigma \in [0, 1]$. ■

571 This is a standard result that is well established in the literature that the optimal environmental
 572 regulation is more stringent when the incumbent government does not give in to the producer lobby's
 573 demand for deregulation. The incumbent government values the welfare of the general voters and
 574 addresses the problem of negative externality from pollution by choosing tougher regulations. Another
 575 takeaway is that the political equilibrium policy outcome converges to the no-lobby strategic equilib-
 576 rium policy outcome as $\lambda \rightarrow \infty$. [Table 1](#) illustrates the results.

577

578 [Insert [Table 1](#) here.]

579

580 I now decompose the total wedge in policies under different regimes.

581 **Corollary 3.** *The total wedge between the global social planner's equilibrium policy and the non-cooperative*
 582 *political equilibrium can be decomposed into the wedge due to non-cooperation with no lobby and the wedge*
 583 *attributed to lobbying.*

$$t^{SW} - t^P = \underbrace{(t^{SW} - t^{ST})}_{\text{non-cooperation wedge with no lobby}} + \underbrace{(t^{ST} - t^P)}_{\text{wedge attributed to lobbying}} \quad (5.25)$$

584 Each of these policy variables is governed by different threshold values of θ .

585 i. The lobbying wedge is positive

$$t^{ST} - t^P = \frac{32(1-c)(2+\sigma)}{D(\sigma, \theta)[\lambda D(\sigma, \theta) + 8]} > 0.$$

586 for every $c \in (0, 1)$; $\sigma, \theta \in [0, 1]$ and $\lambda > 0$.

587 ii. The non-cooperative no-lobby wedge is given by

$$t^{SW} - t^{ST} = \frac{(1-c)(2+\sigma)^2(1-4\theta-\sigma)}{(3+\sigma)D(\sigma, \theta)}$$

588 is positive if and only if $\theta > \tilde{\theta} \equiv \frac{1-\sigma}{4}$.

iii. The total effect is given by

$$t^{SW} - t^P = -\frac{(1-c)(2+\sigma)[\lambda\{N(\sigma, \theta) + 4\} + 8]}{(3+\sigma)[\lambda D(\sigma, \theta) + 8]}$$

. If $\theta \geq \hat{\theta} \equiv \frac{1-\sigma}{4} \forall \lambda > 0, t^{SW} > t^P$. If $\theta < \hat{\theta} \equiv \frac{1-\sigma}{4}, t^{SW} > t^P$ only if $\lambda < \tilde{\lambda}(\sigma, \theta) \equiv -\frac{8}{N(\sigma, \theta) + 4}$.

iv. The threshold value of θ in (ii) is identical to the threshold value in (iii) ($\tilde{\theta} = \hat{\theta}$).

Proof. Given $D(\sigma, \theta) < 0$ and $\lambda D(\sigma, \theta) + 8 < 0$ following the second-order conditions, their product is positive. The numerator is also positive given $c \in (0, 1)$. This implies $t^{ST} > t^P$, which proves (i). In (ii), $t^{SW} > t^{ST}$ if and only if $\theta > \tilde{\theta}(\sigma) \equiv \frac{1-\sigma}{4}$. For (iii), $t^{SW} > t^P$ when $[\lambda N(\sigma, \theta) + 4] + 8 > 0$. Two sub-cases are considered here: (a) if $N(\sigma, \theta) + 4 \geq 0$, which requires $\theta > \tilde{\theta}(\sigma), t^{SW} > t^P$ for every λ ; (b) if $N(\sigma, \theta) + 4 < 0$ for $\theta < \tilde{\theta}(\sigma)$, the negative value of $N(\sigma, \theta) + 4$ becomes more negative as λ rises. To ensure $t^{SW} > t^P$ it requires that the magnitude of the negative term is outweighed by a positive number 8. This yields a threshold value of $\lambda < \tilde{\lambda}(\sigma, \theta) = -\frac{8}{N(\sigma, \theta) + 4} > 0$ for $t^{SW} > t^P$ to hold. Part (iv) shows that the threshold value of $\theta, \hat{\theta}(\sigma) = \frac{2-\sigma-\sigma^2}{4\sigma+8} = \frac{1-\sigma}{4}$ is identical to $\tilde{\theta}$ in (ii). While the lobbying wedge is strictly positive for every θ, λ, σ , the non-cooperation wedge is non-negative if and only if $\theta > \tilde{\theta}$. This yields a sum, which is positive, that is, $(t^{SW} > t^P)$. The condition $\theta < \tilde{\theta}(\sigma)$ does the same job as $\lambda < \tilde{\lambda}(\sigma, \theta)$ [see [Appendix B](#) for detailed proof].

■

5.4.3 Regulation under Product Differentiation

I now illustrate how the degree of product differentiation governs the optimal policy chosen under: (a) global welfare maximisation by the social planner ; (b) national welfare maximisation with strategic trade, non-cooperation, without lobbying; and (c) the political equilibrium with lobbying.

Differentiating t^{SP}, t^{ST} , and t^P with respect to σ yields

i.

$$\frac{dt^{SW}}{d\sigma} = -\frac{1-c}{(3+\sigma)^2} < 0, \quad (5.26)$$

$\forall \sigma \in [0, 1]$ and $c \in (0, 1)$.

ii.

$$\frac{dt^{ST}}{d\sigma} = -\frac{4(1-c)(\sigma^2 + 4\sigma + 8)}{[D(\sigma, \theta)]^2} < 0, \quad (5.27)$$

$\forall \sigma \in [0, 1], c \in (0, 1)$, with $D(\sigma, \theta) < 0$ from the second-order condition which implies $[D(\sigma, \theta)]^2 > 0$,

iii.

$$\frac{dt^P}{d\sigma} = \frac{4\lambda(1-c)[8 - \lambda(\sigma^2 + 4\sigma + 8)]}{[\lambda D(\sigma, \theta) + 8]^2}. \quad (5.28)$$

with $\lambda D(\sigma, \theta) + 8 < 0$ from the second-order condition which implies $[\lambda D(\sigma, \theta) + 8]^2 > 0$,

This is summarised in the following proposition.

Proposition 5. *The relationship between the optimal regulation and product differentiation is non-monotonic in the presence of lobbying.*

i. If $\lambda > \lambda_P(\sigma)$, $\frac{dt^P}{d\sigma} < 0$.

ii. For $\lambda < \lambda_P(\sigma)$, $\frac{dt^P}{d\sigma} > 0$.

iii. $\frac{dt^P}{d\sigma} = 0$ when $\lambda = \lambda_P(\sigma)$.

where $\lambda_P(\sigma) = \frac{8}{\sigma^2 + 4\sigma + 8} \in [\frac{8}{13}, 1] \subset (0, \infty)$.

Otherwise, environmental stringency rises with product differentiation for the global social planner and non-cooperative strategic trade and no-lobby equilibrium.

Proof. Part (i) is straightforward. The derivative is strictly negative as $c \in (0, 1)$ and $\sigma \in [0, 1]$. For part (ii), $\frac{dt^{ST}}{d\sigma} = \frac{(1-c)[N'(\sigma; \theta)D(\sigma, \theta) - D'(\sigma; \theta)N(\sigma, \theta)]}{[D(\sigma, \theta)]^2}$. $N'(\sigma, \theta)D(\sigma, \theta) - D'(\sigma, \theta)N(\sigma, \theta) = -4(\sigma^2 + 4\sigma + 8)$, which is negative for every $\sigma \in [0, 1]$. The product of this term with $(1 - c)$; $c \in (0, 1)$ gives the numerator of the derivative, which is again negative. The denominator is strictly positive as $D(\sigma, \theta) < 0$ following the second-order condition.

In part (iii), $\frac{dt^P}{d\sigma} = \frac{(1-c)[\lambda\{N'(\sigma; \theta)D(\sigma, \theta) - D'(\sigma; \theta)N(\sigma, \theta)\} + 8\{N'(\sigma; \theta) - D'(\sigma; \theta)\}]}{[\lambda D(\sigma, \theta) + 8]^2}$. The numerator simplifies to $4\lambda(1 - c)[8 - \lambda(\sigma^2 + 4\sigma + 8)]$ and the denominator is strictly positive, given the second-order condition $[D(\sigma, \theta) + 8] < 0$ holds. The sign of the derivative is now governed by λ . Defining a threshold value of λ as $\lambda_P(\sigma) = \frac{8}{\sigma^2 + 4\sigma + 8} \in [\frac{8}{13}, 1]$, decreasing in $\sigma \in [0, 1]$ and with $\lambda_P(0) = 1$ and $\lambda_P(1) = \frac{8}{13}$ one can trace out the association between product differentiation and stringency in regulations.

If $\lambda < \lambda_P(\sigma) = \frac{8}{13}$, both the numerator and the denominator are positive, implying $\frac{dt^P}{d\sigma} > 0$. This result flips the differentiation-stringency relationship as in parts (i) and (ii) ($\lambda > \lambda_P(\sigma) = 1$). This implies that as the product becomes more homogeneous ($\sigma \rightarrow 1$), environmental stringency rises, which is a deviation from the generic market structure effect particularly in a Cournot set-up with product differentiation as in scenarios with non-cooperative strategic trade with no lobbying and even the cooperative global social planner's equilibrium policy. For $\lambda \in (\frac{8}{13}, 1)$, there is a non-monotonic relationship between tax and product differentiation. It initially increases with σ , reaches a maximum at $\lambda = \lambda_P(\sigma)$, and then falls [see [Appendix D](#) for detailed proof].

■

This can be explained as follows: For any incumbent government who places at least as much weight on the welfare of the general voters as on the lobby's contribution,² the equilibrium environmental tax is increasing in the degree of product differentiation. As markets become more independent (firms produce highly differentiated or niche products), the stringency of environmental regulations tightens. This result echoes studies including [Fujiwara \(2009\)](#) and [Gautier \(2017\)](#) where the optimal environmental policies for differentiated oligopolies are more stringent. However, there are some structural

²The incumbent government is politically motivated but not purely corrupt and still cares for overall welfare of citizens.

differences driving this result, namely, the presence of producer lobbies and the politically-motivated incumbent. At first glance, it seems counter-intuitive as to why the firm lobby would be content with stringent regulations in the first place. Stringent regulations would raise production costs, thereby adversely affecting the competitive edge of the home firm.

One channel that explains this result is the market power of the domestic firm. With highly differentiated products, the domestic firm acts almost like a monopolist. The product variety is extremely niche that rules out competition from substitutes. As a result the domestic demand becomes highly inelastic. The firm can easily transfer the burden of higher costs due to regulation on the consumers as there is no threat of losing customers. As an offshoot of this channel is that the domestic firm's market share is not harmed. With high product differentiation, the rival firm in the foreign country cannot gain consumers as the variety is unique.

The domestic lobby may support higher environmental regulations as market competition is now geared more towards product attributes and less towards prices (more non-price competition). A niche variety of the product that meets high environmental regulations gains credibility among consumers and sends a strong signal about environmental awareness compared to the product produced by the foreign rival.

Another argument that makes the domestic lobby less averse to stricter environmental regulations is that the loss in marginal profit from higher costs is tempered as the product becomes more differentiated. The niche variety works as an alternative tool for protection, which undermines the need to lower the costs of regulation.

I now consider parts (ii) and (iii). Here, the environmental tax rises with a rise in σ , that is, a fall in the degree of product differentiation $\forall \lambda \leq \lambda_P(\sigma)$. When σ rises, the market becomes more competitive and competing firms may start undercutting prices to capture the market share. No firm will be willing to follow a collusive price as both have an incentive to deviate to absorb additional profit. Industry supply increases and the firm's profit margins fall as they need to sell at lower prices. To insulate the firm from erosion of profits, the incumbent government imposes a higher environmental tax. This may appear as counter-intuitive at the first glance, but this effect can be explained by tax cartelisation. As the environmental tax rises in a market producing homogeneous goods, the firm faces higher costs which is reflected in lower output levels. Lower output levels prevent the price from falling in the market, which protect the profit margin of the firm from falling. The higher tax works as a safety net for the firm when competition rises due to a fall in product differentiation. With a lower λ , the incumbent government cares more about the firm's profits than consumer surplus and environmental quality. The gain in profits due to less competition outweighs the profit lost owing to the rise in the tax. The tax prevents the firm from engaging in price wars, which plummets prices. Although the cost of production rises, the firm is able to maintain higher prices with lower output.

6 Incumbents with import aversion

In this section, I unpack the factors that correlate with protectionist policies when there exists production externalities and the associated environmental regulations. There is evidence that developed countries have resorted to stricter environmental laws as substitutes to trade barriers when the World Trade Organisation (WTO) regulations of liberal trade policies are in place.³ Hence, these ‘ostensible’ measures designed to ensure safety to human life have severe implications on trade flows. Nations which endorse and exercise stringent environmental safety standards limit market access to the exporters around the world, thereby jeopardising their trade competitiveness. The interesting aspect here is that such safety standards work as trade restriction measures. This forms the standard argument for protectionist policies. Protectionist policies also find support from the literature on the canonical framework on lobbying by both polluting firms and green lobbies (Bombardini, 2008; Conconi, 2003; Fredriksson, 1999; Gawande and Bandyopadhyay, 2000; Grossman and Helpman, 1994,9; Maggi and Rodriguez-Clare, 1998). In fact, firms investing in cleaner abatement technology may lobby for strengthening environmental regulations to drive out its rivals who do not make clean investment (Grey, 2018). This result is driven by heterogeneity in production costs and firm productivity (Macedoni and Weinberger, 2024). Even in case of the strategic setting of Cournot duopolies, environmental tax becomes an instrument to alter relative costs and market shares as both domestic and foreign outputs are interdependent (Bontems et al., 2025).

However, empirical evidence on protectionism also links non-economic factors that are politically salient and associated with domestic dominance and foreign penetration. Mayda and Rodrik (2005) suggest that protectionist policies correlate with attributes such as salient status-quo, attitude, national sentiments, prestige, values, and community attachments that affect preferences towards domestic strength and foreign exposure.⁴ The microfoundations of non-economic motives anchored in salient status-quo-based preferences have been provided by Akerlof and Kranton (2000). They model the utility function with attributes such as self image, which is tied to the social status, and a loss from deviations from the salient status-quo-consistent behaviour, also defined as ideal prescriptions. In contrast to Akerlof and Kranton (2000), there could be reference-dependent preferences such as a salient status-quo benchmark policy and any deviation from the exogenous target or reference point is costly to the incumbent.

Drawing motivation from this literature, Unlike Akerlof and Kranton (2000), I allow the incumbent government to derive psychological utility from domestic output dominance relative foreign import penetration. The incumbent government associates a salient status-quo-consistent benchmark level of import, say, $y^0 \geq 0$, and suffers utility losses if the foreign import penetration exceeds this exogenous

³The ban on tuna imports from Mexico by the US government in the year 1990 is an example of how developed nations use (abuse) surrogate policies such as environmental norms to restrict trade. Later in 2001, France imposed an import ban on asbestos and asbestos-containing products citing risks on human health. Recently, the European Union (EU) extended the Emissions Trading Scheme (ETS) on foreign airlines flying into the European airspace. This policy of 2012 received striking opposition from its trading partners like the US, China, India and Russia (Wu and Salzman, 2013).

⁴The ‘America First’ policy championed by Donald Trump is a prominent example where national interests, self sufficiency, sovereignty, and strict immigration policies take precedence over cosmopolitanism and open market access.

714 benchmark/ideal level of imports. The utility loss (disutility) term is given by

$$\psi(y) = -\gamma [\max\{0, y - y^0\}]^2 \quad (6.1)$$

715 which is added to the payoff function of the incumbent government in (5.3). $\gamma > 0$ is the weight
 716 placed on the disutility from benchmark non-conformity. It captures the incumbent's sensitivity towards
 717 any reference dependent policy (here, imports). When $y > y^0$, the disutility is strictly positive and any
 718 $y \leq y^0$ gives zero disutility. If $y^0 = 0$, the incumbent government is a strong protectionist whose
 719 ideal preference is no trade and zero foreign presence. When y^0 reflects imports associated with free
 720 trade, the incumbent government is considered to be anti-protectionist. In a differentiated Cournot
 721 duopoly setting, where domestic and foreign outputs are strategically dependent, the environmental
 722 tax raises domestic firm's cost and benefits the foreign rival firm due to the relative cost advantage. The
 723 environmental tax serves as an instrument to regulate this relative difference to safeguard the domestic
 724 market dominance of the home variety. The incumbent government earns political rent from the Home
 725 firm's dominance over the Foreign firm in the domestic market. Therefore, the incumbent government
 726 uses the tax as a tool to not only contain environmental damage but also cater to the domestic lobby's
 727 demand, and address strategic trade concerns and foreign import aversion.

728 6.1 Incumbent's Payoff

729 With the introduction of import aversion and sovereignty boost associated with a policy, in this case an
 730 environmental tax, the incumbent's payoff function becomes

$$G^I = C(t; t^*) + \lambda G(t; t^*) - \gamma [\max\{0, y - y^0\}]^2; \lambda \in (0, \infty). \quad (6.2)$$

731 It must be noted that the import aversion payoff is associated with the incumbent government
 732 only. Consumers and lobbies have selfish interests, viz., consumers only care about consumer surplus
 733 and lobbies care about profits net of contributions. To be more precise, while consumers want cheaper
 734 goods and a cleaner environment, the lobby prefers laxer regulations to facilitate and safeguard the cost
 735 advantage over its rival. Despite concerns over the average welfare of the voters and the lobbies in-
 736 terests, he/she gets an additional payoff from establishing domestic market dominance, which is tied
 737 to the national sentiment, and it depends on the choice of the environmental tax. The optimal envi-
 738 ronmental tax may not necessarily correct for the environmental externality emanating from pollution.
 739 Therefore, the opportunistic incumbent government can popularise the narrative of national interests
 740 by maintaining a domestic output dominance at the cost of limiting access to the variety of goods for
 741 consumers. Similarly, the environmental tax may be set strategically, which would not reflect true social
 742 damage. Consequently, it reinforces and amplifies the distortion induced by the lobby's influence.

743 Solving for equilibrium environmental tax for $y > y^0$ yields

$$t^I = t^{*I} = \frac{(1-c)[\lambda N(\sigma, \theta) + 8 + 2\gamma\sigma] - 2\gamma\sigma y^0(2+\sigma)}{\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma}. \quad (6.3)$$

Appendix C gives detailed proof.

Since t^I is associated with imports strictly greater than the status-quo benchmark y^0 , the equilibrium tax can be expressed as

$$t^I = \begin{cases} \frac{(1-c)[\lambda N(\sigma, \theta) + 8 + 2\gamma\sigma] - 2\gamma\sigma y^0(2+\sigma)}{\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma} & \text{if } y^0 < y^I \\ t^P & \text{otherwise.} \end{cases} \quad (6.4)$$

The second-order condition gives the upper bound of the national sentiment intensity parameter, γ as

$$\gamma < \bar{\gamma} \equiv -\frac{\lambda D(\sigma, \theta) + 8}{2\sigma} > 0, \quad (6.5)$$

as $\lambda D(\sigma, \theta) + 8 < 0$ from the second-order condition of the political equilibrium. Therefore, $\gamma \in (0, \bar{\gamma})$.

For $y \leq y^0$, the psychological loss goes to zero and $t^I = t^P$. For $y > y^0$, under symmetry, the baseline import penetration level in the political equilibrium is obtained from the equilibrium level of imports, $\bar{y}$. This is the quantity of foreign goods entering the domestic market if the politician did not care about import aversion. It is given by

$$\bar{y} \equiv \frac{1-c-t^P}{2+\sigma} > y^0, \quad (6.6)$$

which is equivalent to the condition $y > y^0$. $\bar{y}$ is strictly positive under the second-order condition $\lambda D(\sigma, \theta) + 8 < 0$.

Plugging in the expression for t^P , gives

$$y^0 < \bar{y} \equiv -\frac{4\lambda(1-c)}{\lambda D(\sigma, \theta) + 8} \quad (6.7)$$

The expression is strictly positive as both the numerator ($-4\lambda(1-c)$) and the denominator ($\lambda D(\sigma, \theta) + 8 < 0$) are negative.

Next, I compare the equilibrium tax under the political equilibrium with that of the national sentiment motive.

Proposition 6. Under $c \in (0, 1)$, $\sigma \in [0, 1]$, $\theta \in [0, 1]$, $\lambda \in (0, \infty)$, and the second-order conditions $\lambda D(\sigma, \theta) + 8 < 0$ and $\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma < 0$, the politician motivated by national sentiments ($y > y^0$) depresses the equilibrium policy relative to the baseline policy chosen in the political equilibrium ($t^I < t^P$) if and only if $y^0 < \bar{y}$.

If $y^0 \geq \bar{y}$, the psychological cost is zero and $t^I = t^P$.

Proof. Using expressions for t^I and t^P one can find $t^P - t^I = \frac{(1-c)[\lambda N(\sigma, \theta) + 8]}{\lambda D(\sigma, \theta) + 8} - \frac{(1-c)[\lambda N(\sigma, \theta) + 8 + 2\gamma\sigma] - 2\gamma\sigma y^0(2+\sigma)}{\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma}$, which simplifies to $t^P - t^I = \frac{2\gamma\sigma(2+\sigma)(y^0 - \bar{y})}{\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma}$. The sign of the denominator is negative under the second-order condition. The sign of the numerator is governed solely

by the term $y^0 - \bar{y}$ as $2\gamma\sigma(2 + \sigma) \geq 0$. For $y^0 < \bar{y} \equiv -\frac{4\lambda(1-c)}{\lambda D(\sigma, \theta) + 8}$, $t^I < t^P$. For $y^0 > \bar{y}$, equilibrium tax rate is t^P . ■

The proposition shows that when the baseline imports exceed the salient status-quo level, that is, $\bar{y} > y^0$, the politician motivated by import aversion chooses a tax that is strictly less than the benchmark level given by the political equilibrium. A reduction in the tax rate improves the competitiveness of the domestic firm due to lower costs. The Cournot framework ensures that imports fall to the acceptable benchmark y^0 as a result of deregulation of the environmental policy. The politician favours the domestic firm over environmental damage.

With the result $t^I < t^P$ if and only if $y^0 < \bar{y}$, one cannot determine the sign of t^I . The sign of t^I can be derived using certain parametric restrictions and regularity conditions.

Since the denominator of $t^I = \frac{(1-c)[\lambda N(\sigma, \theta) + 8 + 2\gamma\sigma] - 2\gamma\sigma y^0(2 + \sigma)}{\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma}$ is negative from the second-order condition, the sign of t^I is governed by the numerator. This leads to the following Proposition:

Proposition 7. *The equilibrium policy under import aversion is*

- i. *Strictly positive ($t^I > 0$) if and only if $t^P > 0$ which requires $\lambda N(\sigma, \theta) + 8 < 0$ and the status quo level of imports y^0 lies in the interval $(\underline{y}, \bar{y})$ which is a non-empty set.*
- ii. *$t^I = t^P > 0$ if and only if $t^P > 0$ and import tolerance is slack such that $y^0 \geq \bar{y}$.*
- iii. *Non-positive such that $t^I \leq 0 < t^P$ indicating a subsidy if and only if $t^P > 0$ and import tolerance being stricter such that $y^0 \leq \underline{y}$.*
- iv. *Strictly negative ($t^I < t^P \leq 0$) if and only if $t^P \leq 0$ (requiring $\lambda N(\sigma, \theta) + 8 \geq 0$) and the import tolerance $y^0 < \bar{y}$ in which the interval $(\underline{y}, \bar{y})$ collapses.*
- v. *$t^I = t^P \leq 0$ if and only if $t^P \leq 0$ and import tolerance is slack such that $y^0 \geq \bar{y}$.*

Proof. Monotonicity condition: Differentiating t^I with respect to y^0 gives $\frac{\partial t^I}{\partial y^0} = -\frac{2\gamma\sigma(2 + \sigma)}{\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma}$, which is strictly positive. t^I is strictly increasing in y^0 with no local minima and maxima.

Threshold values : For t^I to be positive, the numerator $(1 - c)[\lambda N(\sigma, \theta) + 8 + 2\gamma\sigma] - 2\gamma\sigma y^0(2 + \sigma) < 0$, given the second-order condition $\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma < 0$. This gives the condition $y^0 > \underline{y} \equiv \frac{(1-c)[\lambda N(\sigma, \theta) + 8 + 2\gamma\sigma]}{2\gamma\sigma(2 + \sigma)}$, where $\underline{y}$ is the critical import tolerance benchmark at which $t^I = 0$. $\underline{y}$ is the lower bound of the status-quo level of imports y^0 when $t^I > 0$. It is the point where the protectionist policy shifts from a positive tax to a positive subsidy.

The politician with import aversion chooses a strictly lower tax in comparison to the baseline import penetration level in the political equilibrium ($t^I < t^P$) when the baseline imports without any national sentiment $\bar{y}$ exceeds the status-quo level of imports y^0 . This is the upper bound of the status-quo level of imports. Stacking the upper and the lower bounds together gives $\underline{y} < y^0 < \bar{y}$ for a positive tax $t^I > 0$.

Non-emptiness: The interval $(\underline{y}, \bar{y})$ is non-empty and meaningful when the lower bound is strictly less than the upper bound, that is, $\underline{y} < \bar{y}$. For $\underline{y} \geq \bar{y}$, the interval is an empty set and it becomes impossible for t^I to be positive for any y^0 .

Non-emptiness implies that $\bar{y} - \underline{y} > 0$. Considering $\bar{y} - \underline{y} = -\frac{4\lambda(1-c)}{\lambda D(\sigma, \theta) + 8} - \frac{(1-c)[\lambda N(\sigma, \theta) + 8 + 2\gamma\sigma]}{2\gamma\sigma(2+\sigma)} = -\frac{(1-c)[\lambda N(\sigma, \theta) + 8][\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma]}{2\gamma\sigma(2+\sigma)[\lambda D(\sigma, \theta) + 8]}$ to be greater than zero requires $\lambda N(\sigma, \theta) + 8 < 0$, given that the parametric restrictions on c, σ, γ hold along with the second-order conditions $\lambda D(\sigma, \theta) + 8 < 0$ and $\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma < 0$. Therefore, the interval $(\underline{y}, \bar{y})$ is non-empty if and only if $\lambda N(\sigma, \theta) + 8 < 0$, which from the expression of $t^P = \frac{(1-c)[\lambda N(\sigma, \theta) + 8]}{\lambda D(\sigma, \theta) + 8}$ with $c \in (0, 1)$ and the second-order condition $\lambda D(\sigma, \theta) + 8 < 0$ entails the condition for $t^P > 0$.

Equivalent expressions can be derived for $t^P \leq 0$, which requires $\lambda N(\sigma, \theta) + 8 \geq 0$. With $\bar{y} \leq \underline{y}$, any point $y^0 < \bar{y}$ leads to the inequality $y^0 < \underline{y}$. This collapses the interval $(\underline{y}, \bar{y})$ and rules out a positive tax. The monotonicity restriction ensures that $t^I < 0 \forall y^0 < \bar{y}$. With $y^0 \leq \underline{y}$, the monotonicity restriction implies that $t^I(y^0) < t^I(\underline{y}) = 0$. This results in $t^I < t^P \leq 0$, which is a subsidy with active protectionist sentiments. For $y^0 \geq \bar{y}$, protectionist sentiments allow $t^I = t^P \leq 0$.

■

[Proposition 6](#) and [Proposition 7](#) trace out environmental taxation driven by the protectionist sentiments and import tolerance y^0 over a real number line with three mutually exclusive intervals $[0, \underline{y}]$, $(\underline{y}, \bar{y})$ and $[\bar{y}, \infty)$. I consider the case where the politician contains the environmental damage over the firm lobby's pressure for subsidy, that is, $t^P > 0$. When the import tolerance is high such that $y^0 \geq \bar{y}$, national sentiments have nothing much to do as tolerance is very high. The politician sets $t^I = t^P > 0$. When $y^0 \in (\underline{y}, \bar{y})$, national sentiments are activated and the politician deregulates environmental policy crowding out imports. However, the environmental policy is still a positive tax strictly less than the environmental tax in the political equilibrium, that is, $0 < t^I < t^P$. When import tolerance is highly restrictive, such that, $y^0 \leq \underline{y}$, the politician cuts the tax aggressively resulting in a subsidy that gives $t^I \leq 0 < t^P$.

When $t^P \leq 0$, a higher import tolerance $y^0 > \bar{y}$ results in $t^I = t^P \leq 0$ with inactive national sentiments. For $y^0 \leq \bar{y} < \underline{y}$, environmental subsidy deepens ($t^I < t^P \leq 0$) with highly restrictive import tolerance.

[Figure 1](#) and [Figure 2](#) illustrate the import aversion policy outcomes based on the import tolerance y^0 under the political equilibrium tax ($t^P > 0$) and the political equilibrium subsidy ($t^P \leq 0$).

[Insert [Figure 1](#) here.]

[Insert [Figure 2](#) here.]

Following the characterisation of the optimal environmental policy under import aversion and

its relationship with the baseline political equilibrium policy, a natural question that arises is whether the protectionist politician's motive to protect the domestic firm from foreign competition or reducing imports to the target level y^0 come into fruition. More particularly, does the equilibrium tax t^I mitigate the protectionist incumbent's psychological cost, after consideration of the fact that both countries are identical, follow identical logic, identical calculation and move taxes for identical reasons and both react the same way.

Lemma 6.1. *The home country's import function is given by*

$$y = \frac{(1-c)(2-\sigma) + \sigma t - 2t^*}{4-\sigma^2}, \quad (6.8)$$

such that,

$$\frac{\partial y}{\partial t} = \frac{\sigma}{4-\sigma^2} > 0, \quad (6.9)$$

$$\frac{\partial y}{\partial t^*} = -\frac{2}{4-\sigma^2} < 0, \quad (6.10)$$

and

$$\left| \frac{\partial y}{\partial t} \right| < \left| \frac{\partial y}{\partial t^*} \right| \quad \forall \sigma \in [0, 1]. \quad (6.11)$$

Given that the import function is linear in (t, t^*) , for any arbitrary identical combination of the tax (s, s) ,

$$\hat{y} = y(s, s) - y(t, t) = \left(\frac{\partial y}{\partial t} + \frac{\partial y}{\partial t^*} \right) (s - t) = -\frac{1}{2+\sigma} (s - t). \quad (6.12)$$

Proof. Partial differentiation of Home's import function with respect to both t and t^* gives $\frac{\partial y}{\partial t} = \frac{\sigma}{4-\sigma^2} > 0$, and $\frac{\partial y}{\partial t^*} = -\frac{2}{4-\sigma^2} < 0$. Let

$$\begin{aligned} \hat{y} &= y(s, s) - y(t, t) \\ &= \frac{1-c-s}{2+\sigma} - \frac{1-c-t}{2+\sigma} \\ &= \left(\frac{\partial y}{\partial t} + \frac{\partial y}{\partial t^*} \right) (s - t) \\ &= \frac{\sigma-2}{4-\sigma^2} (s - t) \\ &= -\frac{1}{2+\sigma} (s - t) < 0 \quad \forall s > t. \end{aligned} \quad (6.13)$$

■

This shows that the total effect of a tax cut at Home and Foreign leads to a rise in Home imports. The incumbent government at Home cuts taxes to protect the domestic firm and reducing the psychological cost holding its rival's taxes as given. However, the rival adopts the identical policy of cutting taxes for identical reasons. While the tax cut at Home contracts imports, the foreign country moves its tax identically, which increases Home imports. The magnitude of the former effect is outweighed by the

magnitude of the latter. As a result, the home country is flooded with cheaper goods from the foreign country despite the tax cut at Home. This is self-defeating because the protectionist incumbent's policy choice towards reducing imports towards the target backfires and invariably ends up to higher imports. The is purely driven by the psychological and non-pecuniary behaviour rather than retaliation.

This gives the following proposition:

Proposition 8. *Under symmetry where two protectionist incumbents (in Home and in Foreign) are characterised by an identical sensitivity parameter $\gamma = \gamma^*$, the import benchmark $y^0 = x^{*0}$, and $y^0 < \bar{y}$, the equilibrium import level y^I is strictly greater than the baseline import level y^P , precluding import aversion.*

Applying [Proposition 6](#) and [Lemma 6.1](#), the difference between the equilibrium imports under import aversion and that of the baseline policy without import aversion is given by

$$y(t^I, t^I) - y(t^P, t^P) = \frac{t^P - t^I}{2 + \sigma} > 0, \quad (6.14)$$

Proof. Applying [Lemma 6.1](#), with $s = t^I$ and $t = t^P$, $y(t^I, t^I) - y(t^P, t^P) = -\frac{1}{2+\sigma}(t^I - t^P) > 0$ as $t^I < t^P$ whenever $y^0 < \bar{y}$. Plugging in the equilibrium expressions for t^I and t^P , gives

$$\begin{aligned} y(t^I, t^I) - y(t^P, t^P) &= -\frac{1}{2 + \sigma}(t^I - t^P) \\ &= -\frac{1}{2 + \sigma} \left[\frac{(1 - c)[\lambda N(\sigma, \theta) + 8 + 2\gamma\sigma] - 2\gamma\sigma y^0(2 + \sigma)}{\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma} - \frac{(1 - c)[\lambda N(\sigma, \theta) + 8]}{\lambda D(\sigma, \theta) + 8} \right] \\ &= -\frac{1}{2 + \sigma} \left[\frac{(1 - c)(K + 2\gamma\sigma) - 2\gamma\sigma y^0(2 + \sigma)}{R + 2\gamma\sigma} - \frac{(1 - c)K}{R} \right] \end{aligned} \quad (6.15)$$

where $K = \lambda N(\sigma, \theta) + 8$, $R = \lambda D(\sigma, \theta) + 8$, and $K - R = \lambda[N(\sigma, \theta) - D(\sigma, \theta)] = 4\lambda(2 + \sigma)$, and $N(\sigma, \theta) = [\sigma^2 + (4\theta + 1)\sigma + (8\theta - 6)]$ and $D(\sigma, \theta) = \sigma^2 + (4\theta - 3)\sigma + (8\theta - 14)$.

Further simplification gives

$$\begin{aligned} y(t^I, t^I) - y(t^P, t^P) &= -\frac{1}{2 + \sigma} \left[\frac{(1 - c)(K + 2\gamma\sigma)R - 2\gamma\sigma y^0(2 + \sigma)R - (1 - c)K(R + 2\gamma\sigma)}{(R + 2\gamma\sigma)R} \right] \\ &= -\frac{2\gamma\sigma}{2 + \sigma} \left[\frac{(1 - c)(R - K) - y^0(2 + \sigma)R}{(R + 2\gamma\sigma)R} \right] \\ &= -\frac{2\gamma\sigma}{R + 2\gamma\sigma} \left[-\frac{4\lambda(1 - c)}{R} - y^0 \right] \\ &= -\frac{2\gamma\sigma(\bar{y} - y^0)}{\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma} > 0 \quad \forall \quad \bar{y} = -\frac{4\lambda(1 - c)}{\lambda D(\sigma, \theta) + 8} > y^0. \end{aligned} \quad (6.16)$$

[Proposition 8](#) and [Lemma 6.1](#) explain the paradox that the environmental policy (tax/subsidy) chosen by each incumbent government (Home and Foreign) to restrict imports, counter-intuitively expands them when the rival responds symmetrically. The equilibrium import level for an incumbent government with import aversion ⁵ strictly exceeds the import level that would prevail when there is an incumbent government without protectionist sentiment. This property is a result of the Cournot assumption that holds for every $\sigma \in [0, 1]$, regardless of the incumbent's protectionist preferences. The outcome does not rely on the import sensitivity γ or the import target y^0 .

Next, I check how the optimal environmental tax varies with the degree of product differentiation. Differentiating (6.4) with respect to σ yields

$$\frac{dt^I}{d\sigma} = \frac{4\lambda(1-c)[8 - \lambda(\sigma^2 + 4\sigma + 8) - 4\gamma] - 2\gamma y^0 \Phi}{[\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma]^2}, \quad (6.17)$$

where $\Phi = [2(1 + \sigma)(\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma) - \sigma(2 + \sigma)\{\lambda(2\sigma + 4\theta - 3) + 2\gamma\}]$.

This can be stated by the following Proposition:

Proposition 9. *With $c \in (0, 1)$, $\sigma \in [0, 1]$, $\gamma \in [0, \bar{\gamma}]$, $\theta \in [0, 1]$, $y^0 > 0$ and the second-order condition $\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma < 0$, the numerator solely determines the sign of $\frac{dt^I}{d\sigma}$, which is a quadratic in γ , given by $a_2\gamma^2 + a_1\gamma + a_0$, where $a_2 = -4y^0\sigma^2 < 0$, $a_1 = -16\lambda(1 - c) - 2y^0\Phi_0(\sigma, \theta, \lambda)$, $a_0 = 4\lambda(1 - c)[8 - \lambda(\sigma^2 + 4\sigma + 8)]$ and $\Phi_0 = 4\lambda\theta(2 + \sigma)^2 - \lambda(5\sigma^2 + 28\sigma + 28) + 16(1 + \sigma)$. Let $\Delta = a_1^2 - 4a_2a_0 \geq 0$ be the discriminant and $\gamma_1, \gamma_2 \subset [0, \bar{\gamma})$ with $\gamma_1 < \gamma_2$ as the two roots of the quadratic equation in γ .*

i. *The baseline scenario where the politician has no protectionist interests, that is, $\gamma = 0$, the numerator of $\frac{dt^I}{d\sigma}$ is given by a_0 ,*

$$a_0 = \left. \frac{dt^I}{d\sigma} \right|_{\gamma=0} > 0 \text{ if and only if } \lambda < \lambda_P(\sigma) \equiv \frac{8}{\sigma^2 + 4\sigma + 8}.$$

[this follows from [Proposition 5](#)]

ii. *Since $a_2 < 0$, the derivative $\frac{dt^I}{d\sigma}$ is a parabola opening downwards in γ . The sign of the derivative is negative throughout except for a small bounded window (γ_1, γ_2) , where γ_1, γ_2 are the roots of the quadratic equation in γ . Precisely, $\lim_{\gamma \rightarrow \infty} (a_2\gamma^2 + a_1\gamma + a_0) = -\infty$.*

Proof. The proof is trivial. ■

Comparing across different regimes, the prediction of environmental policy change is robust. incumbents cut taxes as market competition intensifies. The result persists across the equilibria, including the global social planner, the strategic trade model, the political lobbying, and the import aversion model. However, the robustness is not independent of some parametric restrictions. The relationship between the tax and the degree of product differentiation reverses whenever the incumbent favours the lobby over aggregate social welfare (low λ).

⁵A robustness check with a linear specification of the psychological cost, given by $\psi(y) = -\gamma y$ is preserved yielding qualitatively similar results. Moreover, the results do not depend on the conditions on y^0 as it does not exist here.

The following proposition addresses the effectiveness of undercutting taxes to restrict foreign imports when actual imports exceed the import tolerance, that is, $\bar{y} > y^0$.

Proposition 10. *When the import aversion binds ($y^0 < \bar{y}$), the symmetric equilibrium import level $y(t^I, t^I)$ is strictly higher than*

i. *the politician's benchmark, that is $y^0 < y(t^I, t^I)$;*

ii. *the non-protectionist benchmark $\bar{y} < y(t^I, t^I)$*

which implies $y^0 < \bar{y} < y(t^I, t^I)$ and $y(t^I, t^I) - \bar{y} = \frac{t^P - t^I}{2 + \sigma} = \frac{2\gamma\sigma[\lambda D(\sigma, \theta) + 8](y^0 - \bar{y})}{[\lambda D(\sigma, \theta) + 8][D(\sigma, \theta) + 8 + 2\gamma\sigma]} > 0$

Proof. From Proposition 7, $t^I < t^P$ iff $y^0 < \bar{y}$. Considering the symmetric equilibrium, $y(t, t) = \frac{1-c-t}{2+\sigma}$ is linear and decreasing in the tax rate. With $t^I < t^P$, one gets $y(t^I, t^I) > y(t^P, t^P) = \bar{y}$. The magnitude of the difference in import levels is given by $y(t^I, t^I) - \bar{y}$. Using the expression for $y(t, t) = \frac{1-c-t}{2+\sigma}$ yields $y(t^I, t^I) - \bar{y} = \frac{t^P - t^I}{2 + \sigma}$. Plugging in expressions for t^P and t^I gives $\frac{t^P - t^I}{2 + \sigma} = \frac{2\gamma\sigma[\lambda D(\sigma, \theta) + 8](y^0 - \bar{y})}{[\lambda D(\sigma, \theta) + 8][D(\sigma, \theta) + 8 + 2\gamma\sigma]}$ which is strictly greater than zero. The denominator has two terms, each being negative following the second-order conditions. The sign of the numerator is determined by the sign of $y^0 - \bar{y}$, given that the second-order condition holds, that is, $\lambda D(\sigma, \theta) + 8 < 0$ and $\gamma \in (0, \bar{\gamma})$ and $\sigma \in [0, 1]$. $y^0 - \bar{y}$ is negative following Proposition 7, which makes $y(t^I, t^I) > \bar{y}$. ■

The equilibrium import level for the home country is given by $y = \frac{(1-c)(2-\sigma)-2t^*+\sigma t}{4-\sigma^2}$. Under symmetry, the home tax has a coefficient $\frac{\sigma}{4-\sigma^2}$ and the foreign tax has a coefficient $-\frac{2}{4-\sigma^2}$. If Foreign raises the tax, the foreign firm's production cost increases and this contracts outputs y and y^* . Owing to Cournot competition between home politician and foreign politician, each of them observes and adjusts its output. When Home's tax rises, it cuts back home output. This induces the rival to expand output y and y^* as the home firm loses to competition from the foreign firm. Notably, Foreign tax has a relatively stronger effect on home imports than Home tax as $\sigma \in [0, 1]$.

The Cournot conjecture suggests that each player chooses its action that maximises its payoff holding the rival's action as given. This yields best response functions for each, which measures the effect of a policy on the players' actions, holding the rival's action as given. In this context,

$$\left. \frac{\partial y}{\partial t} \right|_{\text{constant } t^*} = \frac{\sigma}{4 - \sigma^2} > 0,$$

which shows that a tax cut at Home crowds out inflow of foreign goods. This follows from the first-order condition of each politician's objective function, which boils down to an isolated optimisation problem for each player. Unlike this, there is a comparative static result which highlights the impact on imports driven by changing taxes under the symmetric equilibria. When both politicians act identically

and move taxes, imports rise. The total derivative with taxes moving together is given by

$$\left. \frac{dy}{dt} \right|_{t=t^*} = -\frac{1}{2+\sigma} < 0,$$

. Foreign tax t^* has a relatively stronger effect on home imports than Home tax t as $\sigma \in [0, 1]$. This mimics the standard “race to the bottom” literature where each country’s policy produces desired result while acting individually, holding the rival’s action as given. In contrast, when both countries simultaneously follow identical logic, identical calculation and move taxes for identical reasons, equilibrium outcomes could be the opposite of what any government acting alone would have achieved.

The distinction between a player’s best response while acting individually, holding its rival’s action as given vis-à-vis the total derivative of the symmetric equilibrium with respect to the policy is commonplace in the comparative static literature on oligopoly (Bulow et al., 1985; Dixit, 1986; Seade, 1980).

7 Conclusion

As the world grapples with the acute problem of climate change, globalisation has amplified its impact manifold. With the surge in trade in imperfect markets, sectors producing differentiated products have become significant contributors of greenhouse gas emissions. The present study contributes to the literature on the political economy of environmental policy in a two-country two-firm framework where each firm is a Cournot player producing a differentiated goods and they actively lobby for laxer environmental regulation. The incumbent who values social welfare of the general voters as well as campaign contributions from the lobby, suffers a disutility and a psychological cost tied to foreign penetration.

The central finding of this paper is that product differentiation serves as a facilitator of environmental deregulation when the incumbent exhibits a strong aversion towards foreign penetration. This result is driven by identical incentives of the trading partners, that is, restricting import penetration in the symmetric equilibrium. In isolation, each incumbent elicits an individually rational response to rising imports by cutting taxes. Therefore, any unilateral tax cut to restrict imports at Home is mirrored by an equivalent cut abroad, resulting in higher imports, which are strictly above the incumbent’s own benchmark as well as the import level prevailing under the political equilibrium, absent the psychological cost. This self-defeating protectionist paradox amplifies with product differentiation because of low cross-price elasticity between domestic and foreign goods. Consequently, aggressive tax cuts to restrict imports end up in higher foreign penetration. The outcome is strictly worse than a scenario where the incumbent does nothing at all.

This study also highlights the role of product differentiation in a standard lobbying framework, without import aversion. High product differentiation strengthens the regulatory stringency due to market independence, which insulates domestic producers from foreign cost shocks. Since product

differentiation creates a niche market for differentiated products, the price elasticity of demand falls. This allows firms to readily pass regulation costs onto consumers to keep the market share intact. As a result, firm lobbies do not oppose stricter environmental regulations, inverting the standard prediction that firm lobbying leads to deregulation. On the contrary, as goods become more homogeneous, market competition rises, which pushes the incumbent towards a subsidy instead of a tax.

Drawing comparisons of the above outcomes with the normative benchmarks, namely, the regime of the cooperative social planner and the non-cooperative strategic trade regime, suggest that political lobbying and import aversion create a major distortion of environmental regulations, away from the cooperative and the non-cooperative strategic trade outcomes. The global social planner internalises transboundary externalities and sets a strictly positive tax, regardless of how the damage is shared by the trading partners. The non-cooperative regime without political lobbying differs from the cooperative benchmark as the optimal environmental regulation can move in either direction, conditional on the extent of transboundary externality. When transboundary pollution is low, high product differentiation lowers the elasticity of demand of foreign consumers, which elicit stricter regulations relative to the cooperative benchmark to capture foreign rents. Under high pollution spillovers, the non-cooperative tax falls below the cooperative benchmark. Stricter taxation under high spillovers results in emission leakage and this disincentivises the incumbent to raise taxes.

The self-defeating protectionist paradox closely relates to the green policy competition between the United States (US) and the European Union (EU). When the US, under the US Inflation Reduction Act (2022), committed to tax credits on domestic production of clean-energy, the EU retaliated by introducing concessions and subsidies on clean energy transition under the European Commission's Green Deal Industrial Plan (2023) ([Bloomberg Opinion, 2023](#); [De Ville, 2024](#)). The subsidy war is self-defeating as it raises the global cost of green policy transition.

There are few potential extensions that can be explored to study several policy issues related to sectors producing differentiated products, which are polluting in nature. *First*, one can examine whether import aversion is a substitute for firm lobbying for policy deregulation. *Second*, a comparison of how honest incumbents respond to import aversion as opposed to corrupt ones. *Third*, one can model product differentiation as an endogenous choice where firms strategically over-differentiate products to compel incumbents to adopt aggressive deregulation. *Fourth*, alternative policy instruments such as standards and subsidies could be integrated in the model to make welfare comparisons across these instruments. *Fifth*, cost heterogeneity can be introduced between firms producing differentiated products to account for conflicting interests among firm lobbies. *Sixth*, the study can be extended by introducing green lobbies and studying the dynamics of strategic interaction with producer lobbies. *Last*, the self-defeating protectionist paradox may/may not survive if there is asymmetry in trading nations, such as one of the trading partners being a large country and having less to lose from protectionism. Country size may also reflect heterogeneity in the sensitivity parameter of the disutility from foreign penetration. Geography and location may also alter the realised damage from environmental pollution

¹⁰⁰¹ between trading partners. These are some open questions for exploration.

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1121 **Tables and Figures**

Table 1: Comparison of Policy Regimes across Environmental Damage Parameters

Region	Parameter Range	Strategic Trade (t^{ST})	Political Eq. (t^P)	Explanation
Region I	$0 \leq \theta < \theta_P$	Tax ($t^{ST} > 0$)	Tax ($t^P > 0$)	Taxation Both governments impose a tax to extract terms-of-trade gains from foreign buyers.
Region II	$\theta_P < \theta < \theta_{ST}$	Tax ($t^{ST} > 0$)	Subsidy ($t^P < 0$)	Policy Reversal : Under social welfare (t^{ST}), the government taxes to restrict output. But firm lobbying forces the political government to choose a subsidy ($t^P < 0$) much before as compared to the no-lobby strategic trade regime.
Region III	$\theta_{ST} < \theta \leq 1$	Subsidy ($t^{ST} < 0$)	Subsidy ($t^P < 0$)	Subsidy : Low foreign spillover induces the incumbent government government in both regimes to grant a subsidy to protect market share.

Case I: Tax in the political equilibrium ($t^P > 0 \iff \lambda N(\sigma, \theta) + 8 < 0$)

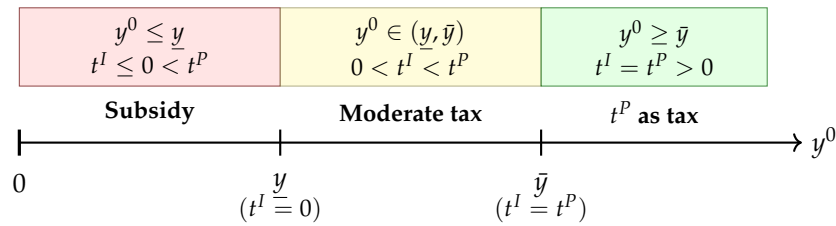


Figure 1: Import tolerance and corresponding policy outcomes under the environmental tax regime ($t^P > 0$)

Case II: Subsidy in the political equilibrium ($t^P \leq 0 \iff \lambda N(\sigma, \theta) + 8 \geq 0$)

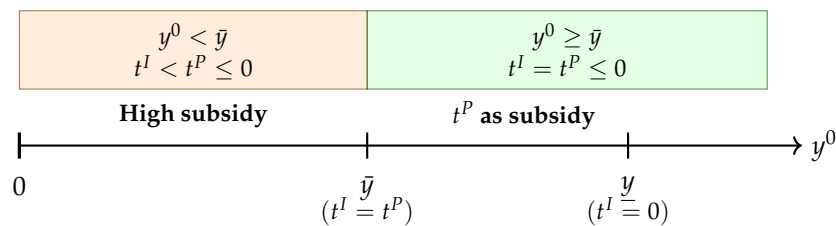


Figure 2: Import tolerance and corresponding policy outcomes under the environmental subsidy regime ($t^P \leq 0$)

Appendix A

Components of the Objective Function $G(t, t^*)$

The aggregate welfare function is given by

$$G(t; t^*) = l + S(t; t^*) + \pi(t; t^*) + R(t; t^*), \quad (\text{A.1})$$

where l is labour income with wage rate equal to unity.

In the following subsections, I derive expressions for each of the components of (A.1).

Consumer Surplus

Consumer surplus is given by

$$S(\cdot) = \frac{1}{2}(x^2 + y^2) + \sigma xy - \left[(1 - \theta) \frac{1}{2}(x + x^*)^2 + \theta \frac{1}{2}(y + y^*)^2 \right]. \quad (\text{A.1})$$

Firm Profits

Profits are given by

$$\pi(\cdot) = (1 - x - \sigma y)x + (1 - x^* - \sigma y^*)x^* - (c + t)(x + x^*). \quad (\text{A.3})$$

Home firm's problem:

$$\underset{x, x^*}{\text{Max}} \quad \pi$$

The first-order conditions are given by

$$\frac{\partial \pi}{\partial x} = x(-1) + (1 - x - \sigma y) - c - t = 0, \quad (\text{A.4})$$

and

$$\frac{\partial \pi}{\partial x^*} = x^*(-1) + (1 - x^* - \sigma y^*) - c - t = 0. \quad (\text{A.5})$$

The first-order conditions can be expressed as:

$$2x + \sigma y = 1 - c - t, \quad (\text{A.6})$$

1135 and

$$2x^* + \sigma y^* = 1 - c - t. \quad (\text{A.7})$$

1136 Foreign firm's problem

$$\underset{y, y^*}{Max} \pi^*$$

1137 The first-order conditions are given by

$$\frac{\partial \pi^*}{\partial y} = y(-1) + (1 - y - \sigma x) - c - t^* = 0, \quad (\text{A.8})$$

1138 and

$$\frac{\partial \pi^*}{\partial y^*} = y^*(-1) + (1 - y^* - \sigma x^*) - c - t^* = 0. \quad (\text{A.9})$$

1139 The first-order conditions can be expressed as:

$$\sigma x + 2y = 1 - c - t^*, \quad (\text{A.10})$$

1140 and

$$\sigma x^* + 2y^* = 1 - c - t^*. \quad (\text{A.11})$$

1141 Solving (A.6) and A.10), and (A.7) and A.11) simultaneously yields

$$x = x^* = \frac{(1 - c)(2 - \sigma) - 2t + \sigma t^*}{4 - \sigma^2}, \quad (\text{A.12})$$

1142 and

$$y = y^* = \frac{(1 - c)(2 - \sigma) + \sigma t - 2t^*}{4 - \sigma^2}. \quad (\text{A.13})$$

1143 Now π and π^* can be expressed as

$$\pi = (1 - x - \sigma y - c - t)x + (1 - x^* - \sigma y^* - c - t)x^*, \quad (\text{A.14})$$

1144 and

$$\pi^* = (1 - y - \sigma x - c - t^*)y + (1 - y^* - \sigma x^* - c - t^*)y^*, \quad (\text{A.15})$$

1145 respectively.

1146 Using (A.6) and (A.7) in (A.14) and (A.10) and (A.11) in (A.15) gives

$$\pi = x^2 + x^{*2}, \quad (\text{A.16})$$

1147 and

$$\pi^* = y^2 + y^{*2}, \quad (\text{A.17})$$

1148 respectively.

1149 **Revenue from Environmental Tax**

1150 Environmental tax revenue is given by

$$R(\cdot) = t(x + x^*). \quad (\text{A.18})$$

1151 **Marginal Effects of a Change in Environmental Tax**

1152 To derive the equilibrium environmental tax, I find the marginal effect of a change in the tax on the
1153 components of the aggregate welfare function in (A.1).

$$\frac{\partial G^P}{\partial t} = \lambda \left[\frac{\partial l}{\partial t} + \frac{\partial S}{\partial t} + \frac{\partial \pi}{\partial t} + \frac{\partial R}{\partial t} \right] + \frac{\partial \pi}{\partial t}. \quad (\text{A.19})$$

1154 where

$$\frac{\partial l}{\partial t} = 0, \quad (\text{A.20})$$

$$\frac{\partial S}{\partial t} = (x + \sigma y) \frac{\partial x}{\partial t} + (y + \sigma x) \frac{\partial y}{\partial t} - \left[(1 - \theta)(x + x^*) \left(\frac{\partial x}{\partial t} + \frac{\partial x^*}{\partial t} \right) + \theta(y + y^*) \left(\frac{\partial y}{\partial t} + \frac{\partial y^*}{\partial t} \right) \right], \quad (\text{A.21})$$

$$\frac{\partial \pi}{\partial t} = 2x \frac{\partial x}{\partial t} + 2x^* \frac{\partial x^*}{\partial t}, \quad (\text{A.22})$$

and

$$\frac{\partial R}{\partial t} = (x + x^*) + t \left(\frac{\partial x}{\partial t} + \frac{\partial x^*}{\partial t} \right). \quad (\text{A.23})$$

Now, the total output at Home and Foreign is given by

$$x + x^* = \frac{1}{4 - \sigma^2} [2(1 - c)(2 - \sigma) - 4t + 2\sigma t^*], \quad (\text{A.24})$$

and

$$y + y^* = \frac{1}{4 - \sigma^2} [2(1 - c)(2 - \sigma) + 2\sigma t - 4t^*]. \quad (\text{A.25})$$

The marginal effect of a change in tax on the change in output is given by

$$\frac{\partial x}{\partial t} = \frac{\partial x^*}{\partial t} = -\frac{2}{4 - \sigma^2} < 0, \quad (\text{A.26})$$

and

$$\frac{\partial y}{\partial t} = \frac{\partial y^*}{\partial t} = \frac{\sigma}{4 - \sigma^2} > 0, \quad (\text{A.27})$$

To find consumer surplus in terms of the environmental tax, I use the following expressions:

$$x + \sigma y = \frac{(1 - c)(2 + \sigma - \sigma^2) - (2 - \sigma^2)t - \sigma t^*}{4 - \sigma^2}, \quad (\text{A.28})$$

and

$$y + \sigma x = \frac{(1 - c)(2 + \sigma - \sigma^2) - \sigma t - (2 - \sigma^2)t^*}{4 - \sigma^2}. \quad (\text{A.29})$$

Using (A.24)-(A.29) in (A.21)-(A.23),

$$\begin{aligned} \frac{\partial S}{\partial t} = \frac{1}{(4 - \sigma^2)^2} & \left[\{ (1 - c)(2 + \sigma - \sigma^2)(\sigma - 2) - (3\sigma^2 - 4)t + \sigma^3 t^* \} \right. \\ & \left. - \{ -4(1 - \theta)(2(1 - c)(2 - \sigma) - 4t + 2\sigma t^*) + 2\sigma\theta(2(1 - c)(2 - \sigma) + 2\sigma t - 4t^*) \} \right], \end{aligned} \quad (\text{A.30})$$

$$\frac{\partial \pi}{\partial t} = -\frac{4}{(4 - \sigma^2)^2} [2(1 - c)(2 - \sigma) - 4t + 2\sigma t^*], \quad (\text{A.31})$$

$$\frac{\partial R}{\partial t} = \frac{1}{4 - \sigma^2} [2(1 - c)(2 - \sigma) - 8t + 2\sigma t^*]. \quad (\text{A.32})$$

Appendix B: Equilibrium tax rates

The Social Planner

The objective function of the social planner is given by

$$\Omega(t, t^*) = G(t, t^*) + G^*(t^*, t). \quad (\text{B.1})$$

The social planner chooses a common tax such that $t = t^*$ internalising the transboundary externality across countries.

Using expressions for $G(t, t^*)$ and $G^*(t^*, t)$,

$$\Omega(t, t^*) = x^2 + y^2 + 2\sigma xy + 2tx + 2ty. \quad (\text{B.2})$$

Using expression (A.30-A.32), the first-order condition is given by

$$2 \left(\frac{(2 - \sigma)(1 - c - t)}{4 - \sigma^2} \right) - \sigma \left(\frac{(2 - \sigma)(1 - c - t)}{4 - \sigma^2} \right) - (2 - \sigma)t = 0, \quad (\text{B.3})$$

which simplifies to

$$(1 - c) - t(3 + \sigma) = 0. \quad (\text{B.4})$$

This yields the equilibrium tax as

$$t^{SW} = \frac{1 - c}{3 + \sigma} \quad c \in (0, 1) \text{ and } \sigma \in [0, 1]. \quad (\text{B.5})$$

The second-order condition is satisfied as

$$\frac{\partial^2 \Omega}{\partial t^2} = -\frac{(2 - \sigma)(3 + \sigma)}{2 + \sigma} < 0 \text{ with } \sigma \in [0, 1]. \quad (\text{B.6})$$

Strategic Trade

The objective function for non-cooperative strategic trade is given by

$$G(t; t^*) = l + S(t; t^*) + \pi(t; t^*) + R(t; t^*). \quad (\text{B.7})$$

1175 Differentiating $G(t; t^*)$ with respect to t with symmetry and then using expressions (A.30-A.32) in the
 1176 above equation and setting it equal to zero yields

$$(1 - c)[- \sigma^2 - (4\theta + 1)\sigma - (8\theta - 6)] + [\sigma^2 + (4\theta - 3)\sigma + (8\theta - 14)]t = 0. \quad (\text{B.8})$$

1177 Solving for t gives

$$t^{ST} = \frac{(1 - c)N(\sigma, \theta)}{D(\sigma, \theta)}, \quad (\text{B.9})$$

1178 where $N(\sigma, \theta) = \sigma^2 + (4\theta + 1)\sigma + (8\theta - 6)$ and $D(\sigma, \theta) = \sigma^2 + (4\theta - 3)\sigma + (8\theta - 14)$. From the second
 1179 order condition, $D(\sigma, \theta) < 0$ for a maximum.

$$\begin{aligned} t^{SW} - t^{ST} &= \frac{1 - c}{3 + \sigma} - \frac{(1 - c)N(\sigma, \theta)}{D(\sigma, \theta)} \\ &= (1 - c) \left[\frac{D(\sigma, \theta) - (3 + \sigma)N(\sigma, \theta)}{(3 + \sigma)D(\sigma, \theta)} \right] \\ &= \frac{(1 - c)(2 + \sigma)^2(1 - 4\theta - \sigma)}{(3 + \sigma)D(\sigma, \theta)}. \end{aligned} \quad (\text{B.10})$$

1180 Political Equilibrium

1181 Assuming symmetry and using (A.30)-(A.32) in (A.19) and setting it equal to zero gives

$$(1 - c)[\lambda\{-\sigma^2 - (4\theta + 1)\sigma - (8\theta - 6)\} - 8] + [\lambda\{\sigma^2 + (4\theta - 3)\sigma + (8\theta - 14)\} + 8]t = 0, \quad (\text{B.11})$$

1182 which yields the equilibrium environmental tax as

$$t^P = \frac{(1 - c)[\lambda N(\sigma, \theta) + 8]}{\lambda D(\sigma, \theta) + 8}. \quad (\text{B.12})$$

1183 The second-order condition requires

$$\frac{\partial^2 G^P}{\partial t^2} = \lambda D(\sigma, \theta) + 8 < 0 \quad \text{with } \sigma \in [0, 1], \theta \in [0, 1]. \quad (\text{B.13})$$

1184 Tax Wedge

1185 The wedge between the global planner's tax t^{SW} and the tax under the political equilibrium t^P can
 1186 be decomposed into two effects, the non-cooperative wedge with no lobby, $t^{SW} - t^{ST}$ and the wedge
 1187 attributed to lobbying, $t^{ST} - t^P$.

$$\begin{aligned}
t^{SW} - t^{ST} &= \frac{(1-c)}{3+\sigma} - \frac{(1-c)N(\sigma, \theta)}{D(\sigma, \theta)} \\
&= (1-c) \left[\frac{D(\sigma, \theta) - (3+\sigma)N(\sigma, \theta)}{(3+\sigma)D(\sigma, \theta)} \right] \\
&= \frac{(1-c)(2+\sigma)^2(1-4\theta-\sigma)}{(3+\sigma)D(\sigma, \theta)}, \tag{B.14}
\end{aligned}$$

1188 where the sign depends on threshold value of θ , that is $\theta_P(\sigma, \lambda)$. Next

$$\begin{aligned}
t^{ST} - t^P &= \frac{(1-c)N(\sigma, \theta)}{D(\sigma, \theta)} - \frac{(1-c)[\lambda N(\sigma, \theta) + 8]}{\lambda D(\sigma, \theta) + 8} \\
&= (1-c) \frac{N(\sigma, \theta)[\lambda D(\sigma, \theta) + 8] - D(\sigma, \theta)[\lambda N(\sigma, \theta) + 8]}{D(\sigma, \theta)[\lambda D(\sigma, \theta) + 8]} \\
&= \frac{32(1-c)(2+\sigma)}{D(\sigma, \theta)[\lambda D(\sigma, \theta) + 8]} > 0. \tag{B.15}
\end{aligned}$$

1189 Appendix C

1190 Equilibrium Environmental Tax under Import Aversion

1191 The incumbent's objective function is given by

$$G^I = C(t; t^*) + \lambda G(t; t^*) - \gamma [\max\{0, y - y^0\}]^2; \lambda \in (0, \infty). \tag{C.1}$$

1192 The objective function of the incumbent government is embedded in the disutility from import penetra-
1193 tion given by

$$\psi(y) = -\gamma [\max\{0, y - y^0\}]^2. \tag{C.2}$$

1194 Now the marginal effect of the tax on the disutility is given by

$$\frac{\partial \psi}{\partial t} = -2\gamma(y - y^0) \frac{\partial y}{\partial t} = -\frac{2\gamma\sigma}{(4 - \sigma^2)^2} [(1-c)(2-\sigma) + \sigma t - 2t^* - y^0(4 - \sigma^2)]. \tag{C.3}$$

1195 Assuming symmetry and using (A.30)-(A.32) in (C.1) including the incumbent's disutility, the first-
1196 order condition is given by

$$\begin{aligned}
(1-c)[\lambda\{-\sigma^2 - (4\theta+1)\sigma - (8\theta-6)\} - 8 - 2\gamma\sigma] + 2\gamma\sigma y^0(2+\sigma) \\
+ [\lambda\{\sigma^2 + (4\theta-3)\sigma + (8\theta-14)\} + 8 + 2\gamma\sigma]t = 0, \tag{C.4}
\end{aligned}$$

1197 which yields the equilibrium environmental tax as

$$\begin{aligned}
t^I &= \frac{(1-c)[\lambda\{\sigma^2 + (4\theta + 1)\sigma + (8\theta - 6)\} + 8 + 2\gamma\sigma] - 2\gamma\sigma y^0(2 + \sigma)}{\lambda\{\sigma^2 + (4\theta - 3)\sigma + (8\theta - 14)\} + 8 + 2\gamma\sigma} \\
&= \frac{(1-c)[\lambda N(\sigma, \theta) + 8 + 2\gamma\sigma] - 2\gamma\sigma y^0(2 + \sigma)}{\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma}.
\end{aligned} \tag{C.5}$$

1198 The second-order condition is given by

$$\frac{\partial^2 G^I}{\partial t^2} = (2 - \sigma)[\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma] < 0. \tag{C.6}$$

1199 Appendix D

1200 Comparative Statics

1201 I derive the effect of product differentiation on the equilibrium taxes, namely, the global social planner's
 1202 tax (t^{SW}), the non-cooperative strategic trade tax (t^{ST}), the political lobbying tax (t^P), and the political
 1203 lobbying under import aversion tax (t^I)

1204 Global social planner benchmark $\left(\frac{dt^{SW}}{d\sigma}\right)$

1205 Given the optimal global social welfare tax rate:

$$t^{SW} = \frac{1 - c}{3 + \sigma}. \tag{D.1}$$

1206 Differentiating directly with respect to σ :

$$\begin{aligned}
\frac{dt^{SW}}{d\sigma} &= (1 - c) \cdot \frac{d}{d\sigma} \left(\frac{1}{3 + \sigma} \right) \\
&= -\frac{1 - c}{(3 + \sigma)^2}.
\end{aligned} \tag{D.2}$$

1207 —

1208 Strategic trade equilibrium $\left(\frac{dt^{ST}}{d\sigma}\right)$

1209 Given the non-cooperative strategic tax rate:

$$t^{ST} = \frac{(1 - c)N(\sigma, \theta)}{D(\sigma, \theta)}, \tag{D.3}$$

1210 where:

$$N(\sigma, \theta) = \sigma^2 + (4\theta + 1)\sigma + (8\theta - 6)$$

1211 ,

$$D(\sigma, \theta) = \sigma^2 + (4\theta - 3)\sigma + (8\theta - 14).$$

1212 and

$$N(\sigma, \theta) - D(\sigma, \theta) = 4(\sigma + 2). \quad (\text{D.4})$$

1213 Now, $N'(\sigma, \theta) = D'(\sigma, \theta) + 4$, where $D'(\sigma, \theta) = 2\sigma + 4\theta - 3$.1214 Differentiating the tax rate with respect to σ gives

$$\begin{aligned} \frac{dt^{ST}}{d\sigma} &= (1-c) \frac{(D' + 4)D - [D + 4(\sigma + 2)]D'}{[D(\sigma, \theta)]^2} \\ &= \frac{4(1-c) [D(\sigma, \theta) - (\sigma + 2)D'(\sigma, \theta)]}{[D(\sigma, \theta)]^2} \\ &= \frac{4(1-c) [(\sigma^2 + (4\theta - 3)\sigma + 8\theta - 14) - (\sigma + 2)(2\sigma + 4\theta - 3)]}{[D(\sigma, \theta)]^2} \\ &= \frac{4(1-c) [\sigma^2 + (4\theta - 3)\sigma + 8\theta - 14 - (2\sigma^2 + (4\theta + 1)\sigma + 8\theta - 6)]}{[D(\sigma, \theta)]^2} \\ &= -\frac{4(1-c)(\sigma^2 + 4\sigma + 8)}{[D(\sigma, \theta)]^2}. \end{aligned} \quad (\text{D.5})$$

1215 **Tax under Political Lobbying** $\left(\frac{dt^P}{d\sigma} \right)$

1216 Given the equilibrium tax

$$t^P = \frac{(1-c)[\lambda N(\sigma, \theta) + 8]}{\lambda D(\sigma, \theta) + 8}. \quad (\text{D.6})$$

1217 Applying the quotient rule with respect to σ :

$$\begin{aligned} \frac{dt^P}{d\sigma} &= (1-c) \left[\frac{\lambda N'(\sigma, \theta)[\lambda D(\sigma, \theta) + 8] - [\lambda N(\sigma, \theta) + 8]\lambda D'(\sigma, \theta)}{[\lambda D(\sigma, \theta) + 8]^2} \right] \\ &= \frac{\lambda(1-c) \{ \lambda [N'D - ND'] + 8[N' - D'] \}}{[\lambda D(\sigma, \theta) + 8]^2} \\ &= \frac{\lambda(1-c) \{ -4\lambda(\sigma^2 + 4\sigma + 8) + 32 \}}{[\lambda D(\sigma, \theta) + 8]^2} \\ &= -\frac{4\lambda(1-c) [\lambda(\sigma^2 + 4\sigma + 8) - 8]}{[\lambda D(\sigma, \theta) + 8]^2}. \end{aligned} \quad (\text{D.7})$$

1218 **Tax under Import Aversion** $\left(\frac{dt^I}{d\sigma} \right)$

1219 Given the baseline policy tax:

$$t^I = \frac{(1-c)[\lambda N(\sigma, \theta) + 8 + 2\gamma\sigma] - 2\gamma\sigma y^0(2 + \sigma)}{\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma} \equiv \frac{N^I}{D^I}, \quad (\text{D.8})$$

1220 where $N^I = (1-c)[\lambda N(\sigma, \theta) + 8 + 2\gamma\sigma] - 2\gamma\sigma y^0(2 + \sigma)$ and $D^I = \lambda D(\sigma, \theta) + 8 + 2\gamma\sigma$.

$$\frac{dt^I}{d\sigma} = \frac{D^I N^{I'} - N^I D^{I'}}{D^{I^2}}, \quad (\text{D.9})$$

where $N^{I'} = 4\lambda(1-c)[8 - \lambda(\sigma^2 + 4\sigma + 8) - 4\gamma] - 2\gamma y^0\Phi$, $D^{I'} = \lambda(2\sigma + 4\theta - 3) + 2\gamma$, and $\Phi = [2(1+\sigma)(\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma) - \sigma(2+\sigma)\{\lambda(2\sigma + 4\theta - 3) + 2\gamma\}]$.

Further simplification and substitution gives

$$\frac{dt^I}{d\sigma} = \frac{4\lambda(1-c)[8 - \lambda(\sigma^2 + 4\sigma + 8) - 4\gamma] - 2\gamma y^0\Phi}{[\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma]^2} \equiv \frac{N_0^I}{D_0^{I'}}, \quad (\text{D.10})$$

where $N_0^I = 4\lambda(1-c)[8 - \lambda(\sigma^2 + 4\sigma + 8) - 4\gamma] - 2\gamma y^0\Phi$ and $D_0^I = [\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma]^2$.

Since the denominator D_0^I is a square of the second-order condition $\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma < 0$, it is positive. So the sign of the derivative solely depends on the sign of the numerator N_0^I , which can be expressed as $A_1 - A_2$. Here $A_1 = 4\lambda(1-c)[8 - \lambda(\sigma^2 + 4\sigma + 8) - 4\gamma]$ and $A_2 = 2\gamma y^0\Phi$.

Isolating terms with γ from A_1 and A_2 yields $A_1 = 4\lambda(1-c)[8 - \lambda(\sigma^2 + 4\sigma + 8)] - 16\lambda\gamma(1-c)$, and $A_2 = 2\gamma y^0\Phi_0 + 4\gamma^2 y^0\sigma^2$.

Now the numerator N_0^I becomes a quadratic in γ , given by $N_0^I = a_2\gamma^2 + a_1\gamma + a_0$, where $a_2 = -4y^0\sigma^2$, $a_1 = -16\lambda(1-c) - 2y^0\Phi_0$, $a_0 = 4\lambda(1-c)[8 - \lambda(\sigma^2 + 4\sigma + 8)]$ and $\Phi_0 = \Phi - 2\gamma\sigma^2$.

Setting $N_0^I = 0$, gives the roots γ_1, γ_2 such that $\gamma_1 < \gamma_2$ of the quadratic equation $a_2\gamma^2 + a_1\gamma + a_0 = 0$. For real roots, the discriminant $\Delta = a_1^2 - 4a_0a_2 \geq 0$. Factorisation gives $N_0^I = a_2(\gamma - \gamma_1)(\gamma - \gamma_2)$ with $a_2 = -4y^0\sigma^2 \leq 0$ as $y^0 \geq 0$, and $\sigma \in [0, 1]$. The quadratic function changes sign for different values of γ . This leads to the result that the relationship of the environmental tax with product differentiation is non-monotone with respect to $\gamma \in [0, \bar{\gamma})$.

Thus, the derivative is:

$$\frac{dt^I}{d\sigma} = \frac{a_2\gamma^2 + a_1\gamma + a_0}{[\lambda D(\sigma, \theta) + 8 + 2\gamma\sigma]^2}. \quad (\text{D.11})$$

Statements and Declarations

Competing Interests: The author certifies that he has no affiliations with or involvement in any organisation or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.